

Page Collection

Bill No.	Description		Amount in figure Rs.
1	Site Clearance and Dismantling	Page No.3	48,42,479
		Page No.4	16,89,230
	Sub Total		65,31,709
2	Earthworks	Page No.5	59,21,80,240
	Sub Total		59,21,80,240
3	Granular Sub-Base and Base Courses	Page No.6	67,53,28,538
	Sub Total		67,53,28,538
4	Bituminous Courses	Page No.7	39,29,40,187
	Sub Total		39,29,40,187
5	Cross Drainage Works(Culverts)	Page No.8	29,16,82,663
		Page No.9	40,16,13,115
	Sub Total		69,32,95,778
6	Bridges/ Underpasses	Page No.10	8,97,49,238
		Page No.11	4,96,824
		Page No.12	3,55,27,993
		Page No.13	61,45,602
	Sub Total		13,19,19,657
7	Repair of Existing Structures	Page No.14	0
	Sub Total		0
8	Drainage and Protective Works	Page No.15	1,09,71,18,826
		Page No.16	1,72,61,86,295
	Sub Total		2,82,33,05,121
9	Traffic Signs, Markings and Road Appurtenances	Page No.17	7,78,91,003
		Page No.18	1,75,25,664
	Sub Total		9,54,16,668
10	Miscellaneous	Page No.19	1,54,44,655
	Sub Total		1,54,44,655

**BILL OF QUANTITIES (BOQ)
BILL NO. 1: GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.01	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	75	58,066	43,48,017
1.02	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.				
	(i) Brick Masonry	cum	Nil	362	-
	(ii) Stone Masonry	cum	945	362	3,42,090
	(iii) Lime/Plain Cement Concrete in all grades	cum	78	559	43,602
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum	30	938	28,140
	(v) Bituminous courses	cum	60	346	20,760
	(vi) Non Bituminous courses	cum	Nil	586	-
	(vii) Cement concrete Pavements	cum	Nil	1,834	-
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.	5	256	1,280
	(ix) Hectometre Stone	No.	Nil	51	-
	(x) 5th KM stone	No.	Nil	438	-
	(xi) Kerb stone	m	Nil	18	-
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m	Nil	55	-
	(xiii) RCC Railing	m	Nil	261	-
	(xiv) Guard Rails/W Beam Crash Barrier	m	Nil	88	-
	(xv) Brick Pitching / Brick soling	cum	Nil	235	-
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and dismantling of foundation concrete and lines under the supervision of concerned department.	m	Nil	195	-
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts including earthwork and dismantling of masonry works.		Nil		-
	(a) Up to 600 mm dia	m	Nil	206	-
	(b) Above 600 mm to 900 mm dia	m	210	279	58,590
	(c) Above 900 mm dia	m	Nil	478	-
	(xviii) Dismantling stone pitching/ dry stone spalls	cum	Nil	251	-
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	m	Nil	147	-
	Carried to Page Collection				48,42,479

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO. 1: GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.03	Cutting of standing trees as per Technical Specification Clause 201				
	i) Girth from 300mm to 600mm	Each	400	59	23,600
	ii) Girth above 600mm to 900mm	Each	500	106	53,000
	iii) Girth above 900mm to 1800mm	Each	600	205	1,23,000
	iv) Girth above 1800mm	Each	500	387	1,93,500
1.04	Removal of Stumps and roots as per Technical Specification Clause 201 and 305.				
	i) Girth from 300mm to 600mm	Each	440	176	77,440
	ii) Girth above 600mm to 900mm	Each	550	319	1,75,450
	iii) Girth above 900mm to 1800mm	Each	660	614	4,05,240
	iv) Girth above 1800mm	Each	550	1,160	6,38,000
	Carried to Page Collection				16,89,230

**BILL OF QUANTITIES (BOQ)
BILL NO.5: CROSS DRAINAGE STRUCTURES (CULVERTS)**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
5.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in				
	(a) All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	17,437	262	45,59,776
	(b) Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	Nil	427	-
	(c) Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	Nil	419	-
	(d) Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	40,686	562	2,28,65,532
5.02	Plain Cement Concrete (M 15 Grade) for Levelling Course below Headwall, Abutment, Wing wall, Around foundation in rock, Approach slab, Chutes, Catch Pits etc. complete as per drawings and Technical Specification Section 1700, 2100, 2200, 2500 and 2700.	cum	1,027	11,345	1,16,51,315
5.03	RCC/PCC for open foundations complete excluding reinforcement as per drawings and Technical Specification Section 1700 and 2100				
	(a) M 25 Grade	cum	Nil	13,493	-
	(b) M 30 Grade	cum	2,633	13,502	3,55,50,766
5.04	RCC / PCC for Substructure, Wing wall, Retaining wall, Box Cell Structures etc. excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2200.				
	(a) M 25 Grade	cum	Nil	14,307	-
	(b) M 30 Grade	cum	6,968	14,350	9,99,90,800
5.05	Coursed Rubble Masonry in 1:3 Cement Mortar for Foundation and Substructure complete as per Drawing and Technical Specification Clause 1400, 2100 and 2200.	cum	Nil	10,745	-
5.06	RCC (M 30 Grade) for Slab Culvert Superstructure excluding reinforcement complete as per drawings and Technical specifications Section 1700 and 2300.	cum	Nil	15,549	-
5.07	Steel Reinforcement (Fe 500D) for Foundations, Substructures and superstructures etc. complete as per drawings and Technical Specification Section 1600.	MT	1,166	91,839	10,70,84,274
5.08	Approach Slab (RCC M 30 Grade) including reinforcement complete as per drawings and Technical Specification Section 2700.	cum	556	17,950	99,80,200
5.09	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	Nil	17,167	-
5.10	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	Nil	3,080	-
Carried to Page Collection					29,16,82,663

BILL OF QUANTITIES
BILL NO. 6: NEW BRIDGES

Item	Description	Unit	Quantity	Unit Rate	Amount (Rupees)
6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000				
a	Tar Paper Bearings	sqm	37	140	5,180
b	Elastomeric Bearings	cu.cm	Nil	7	-
c	POT cum PTEE Bearings		Nil		-
i	Fixed Bearings, having Capacity of		Nil		-
	925 T ± 25 T	nos	Nil	1,61,823	-
	525 T ± 25 T	nos	Nil	91,845	-
	325 T ± 25 T	nos	Nil	29,591	-
	275 T ± 25 T	nos	Nil	25,039	-
ii	Free / Guided Bearings, having Capacity of		Nil		-
	925 T ± 25 T	nos	Nil	2,35,298	-
	825 T ± 25 T	nos	Nil	2,09,861	-
	725 T ± 25 T	nos	Nil	1,84,423	-
	675 T ± 25 T	nos	Nil	1,71,704	-
	575 T ± 25 T	nos	Nil	1,46,267	-
	525 T ± 25 T	nos	Nil	1,33,548	-
	325 T ± 25 T	nos	Nil	41,832	-
	275 T ± 25 T	nos	8	35,396	2,83,168
iii	Pin Bearings, having Capacity of		Nil		-
	50 T ± 12.5 T	nos	Nil	16,783	-
	75 T ± 12.5 T	nos	Nil	25,174	-
	100 T ± 12.5 T	nos	Nil	33,565	-
	125 T ± 12.5 T	nos	Nil	50,682	-
	150 T ± 12.5 T	nos	2	71,289	1,42,578
iv	Metallic Guide Bearings, having Capacity of		Nil		-
	25 T ± 12.5 T	nos	Nil	8,237	-
	50 T ± 12.5 T	nos	Nil	16,475	-
	75 T ± 12.5 T	nos	Nil	24,712	-
	100 T ± 12.5 T	nos	2	32,949	65,898
6.11	Elastomeric Stopper Pad for Seismic Restra ner as per drawings and Technical Specification 2000	cu.cm	Nil	7	-
Carried to Page Collection					4,96,824

ABSTRACT OF QUANTITIES
BILL NO. 6: BRIDGES

Item	Description	Unit	No.	Length (m)	Breadth h (m)	Height / Depth(m)	Quantity
6.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in						
a	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	As per annexure 5.1				3,251
b	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum					-
c	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum					-
d	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	As per annexure 5.1				1,898
6.02	PCC (M 15 Grade) in Levelling course below Foundation, Pile Cap, Approach slab, Annular filling around foundation in rock, etc. complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	As per annexure 5.1				191
6.03	RCC / PCC in open foundations excluding reinforcement complete as per drawings and Technical Specification Sections 1700 and 2100.						-
a	M 30 Grade	cum					-
b	M 35 Grade	cum	As per annexure 5.1				1,275
6.04	RCC / PCC in Substructure, Wing Wall, retaining wall, Box Cell Structures, Seismic restrainer etc. excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200						-
a	M 25	cum					-
b	M 30	cum					-
c	M 35	cum	As per annexure 5.1				1,360
6.05	RCC in Superstructure excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2300.						-
a	Solid Slab (M 30 Grade)	cum	As per annexure 5.1				80
b	Deck Slab for Composite Girder (M 35 Grade)	cum					-
c	Voided Slab (M 35 Grade)	cum	As per annexure 5.1				463
6.06	Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600 .	MT	As per annexure 5.1				421
6.07	Steel Girder for Steel Composite Superstructure including railing and fixing of girder with Bearing complete as per drawings and Technical Specification 1000 and 1900.	MT					-
6.08	Bituminous (Type 2) Wearing Coat as per drawings and Technical Specification Section 2700.	sqm	As per annexure 5.1				621
6.09	40 thk. PCC (M25) finished with 15 thk. Plaster (1:3) complete as per drawings and Technical Specification.	cum	As per annexure 5.1				10

ABSTRACT OF QUANTITIES
BILL NO. 6: BRIDGES

Item	Description	Unit	No.	Length (m)	Breadth h (m)	Height / Depth(m)	Quantity
6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000						-
a	Tar Paper Bearings	sqm	As per annexure 6.1				37
b	Elastomeric Bearings	cu cm					-
c	POT cum PTEE Bearings						-
i	Fixed Bearings, having Capacity of						-
	925 T ± 25 T	nos					-
	525 T ± 25 T	nos					-
	325 T ± 25 T	nos					-
	275 T ± 25 T	nos					-
ii	Free / Guided Bearings, having Capacity of						-
	925 T ± 25 T	nos					-
	825 T ± 25 T	nos					-
	725 T ± 25 T	nos					-
	675 T ± 25 T	nos					-
	575 T ± 25 T	nos					-
	525 T ± 25 T	nos					-
	325 T ± 25 T	nos					-
	275 T ± 25 T	nos					-
iii	Pin Bearings, having Capacity of		As per annexure 6.1				8
	50 T ± 12.5 T	nos					-
	75 T ± 12.5 T	nos					-
	100 T ± 12.5 T	nos					-
	125 T ± 12.5 T	nos					-
	150 T ± 12.5 T	nos					-
iv	Metallic Guide Bearings, having Capacity of		As per annexure 6.1				2
	25 T ± 12.5 T	nos					-
	50 T ± 12.5 T	nos					-
	75 T ± 12.5 T	nos					-
	100 T ± 12.5 T	nos					-
	125 T ± 12.5 T	nos					-
	150 T ± 12.5 T	nos					-
	200 T ± 12.5 T	nos					-
	250 T ± 12.5 T	nos					-
	300 T ± 12.5 T	nos					-
	350 T ± 12.5 T	nos					-
	400 T ± 12.5 T	nos					-
	450 T ± 12.5 T	nos					-
	500 T ± 12.5 T	nos					-
	550 T ± 12.5 T	nos					-
	600 T ± 12.5 T	nos					-
	650 T ± 12.5 T	nos					-
	700 T ± 12.5 T	nos					-
	750 T ± 12.5 T	nos					-
	800 T ± 12.5 T	nos					-
	850 T ± 12.5 T	nos					-
	900 T ± 12.5 T	nos					-
	950 T ± 12.5 T	nos					-
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	1050 T ± 12.5 T	nos					-
	1100 T ± 12.5 T	nos					-
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	1200 T ± 12.5 T	nos					-
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	1350 T ± 12.5 T	nos					-
	1400 T ± 12.5 T	nos					-
	1450 T ± 12.5 T	nos					-
	1500 T ± 12.5 T	nos					-
	1550 T ± 12.5 T	nos					-
	1600 T ± 12.5 T	nos					-
	1650 T ± 12.5 T	nos					-
	1700 T ± 12.5 T	nos					-
	1750 T ± 12.5 T	nos					-
	1800 T ± 12.5 T	nos					-
	1850 T ± 12.5 T	nos					-
	1900 T ± 12.5 T	nos					-
	1950 T ± 12.5 T	nos					-
	2000 T ± 12.5 T	nos					-
	2050 T ± 12.5 T	nos					-
	2100 T ± 12.5 T	nos					-
	2150 T ± 12.5 T	nos					-
	2200 T ± 12.5 T	nos					-
	2250 T ± 12.5 T	nos					-
	2300 T ± 12.5 T	nos					-
	2350 T ± 12.5 T	nos					-
	2400 T ± 12.5 T	nos					-
	2450 T ± 12.5 T	nos					-
	2500 T ± 12.5 T	nos					-
	2550 T ± 12.5 T	nos					-
	2600 T ± 12.5 T	nos					-
	2650 T ± 12.5 T	nos					-
	2700 T ± 12.5 T	nos					-
	2750 T ± 12.5 T	nos					-
	2800 T ± 12.5 T	nos					-
	2850 T ± 12.5 T	nos					-
	2900 T ± 12.5 T	nos					-
	2950 T ± 12.5 T	nos					-
	3000 T ± 12.5 T	nos					-
	3050 T ± 12.5 T	nos					-
	3100 T ± 12.5 T	nos					-
	3150 T ± 12.5 T	nos					-
	3200 T ± 12.5 T	nos					-
	3250 T ± 12.5 T	nos					-
	3300 T ± 12.5 T	nos					-
	3350 T ± 12.5 T	nos					-
	3400 T ± 12.5 T	nos					-
	3450 T ± 12.5 T	nos					-
	3500 T ± 12.5 T	nos					-
	3550 T ± 12.5 T	nos					-
	3600 T ± 12.5 T	nos					-
	3650 T ± 12.5 T	nos					-
	3700 T ± 12.5 T	nos					-
	3750 T ± 12.5 T	nos					-
	3800 T ± 12.5 T	nos					-
	3850 T ± 12.5 T	nos					-
	3900 T ± 12.5 T	nos					-
	3950 T ± 12.5 T	nos					-
	4000 T ± 12.5 T	nos					-
	4050 T ± 12.5 T	nos					-
	4100 T ± 12.5 T	nos					-
	4150 T ± 12.5 T	nos					-
	4200 T ± 12.5 T	nos					-
	4250 T ± 12.5 T	nos					-
	4300 T ± 12.5 T	nos					-
	4350 T ± 12.5 T	nos					-
	4400 T ± 12.5 T	nos					-
	4450 T ± 12.5 T	nos					-
	4500 T ± 12.5 T	nos					-
	4550 T ± 12.5 T	nos					-
	4600 T ± 12.5 T	nos					-
	4650 T ± 12.5 T	nos					-
	4700 T ± 12.5 T	nos					-
	4750 T ± 12.5 T	nos					-
	4800 T ± 12.5 T	nos					-
	4850 T ± 12.5 T	nos					-
	4900 T ± 12.5 T	nos					-
	4950 T ± 12.5 T	nos					-
	5000 T ± 12.5 T	nos					-
	5050 T ± 12.5 T	nos					-
	5100 T ± 12.5 T	nos					-
	5150 T ± 12.5 T	nos					-
	5200 T ± 12.5 T	nos					-
	5250 T ± 12.5 T	nos					-
	5300 T ± 12.5 T	nos					-
	5350 T ± 12.5 T	nos					-
	5400 T ± 12.5 T	nos					-
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	5800 T ± 12.5 T	nos					-
	5850 T ± 12.5 T	nos					-
	5900 T ± 12.5 T	nos					-
	5950 T ± 12.5 T	nos					-
	6000 T ± 12.5 T	nos					-
	6050 T ± 12.5 T	nos					-
	6100 T ± 12.5 T	nos					-
	6150 T ± 12.5 T	nos					-
	6200 T ± 12.5 T	nos					-
	6250 T ± 12.5 T	nos					-
	6300 T ± 12.5 T	nos					-
	6350 T ± 12.5 T	nos					-
	6400 T ± 12.5 T	nos					-
	6450 T ± 12.5 T	nos					-
	6500 T ± 12.5 T	nos					-
	6550 T ± 12.5 T	nos					-
	6600 T ± 12.5 T	nos					-
	6650 T ± 12.5 T	nos					-
	6700 T ± 12.5 T	nos					-
	6750 T ± 12.5 T	nos					-
	6800 T ± 12.5 T	nos					-
	6850 T ± 12.5 T	nos					-
	6900 T ± 12.5 T	nos					-
	6950 T ± 12.5 T	nos					-
	7000 T ± 12.5 T	nos					-
	7050 T ± 12.5 T	nos					-
	7100 T ± 12.5 T	nos					-
	7150 T ± 12.5 T	nos					-
	7200 T ± 12.5 T	nos					-
	7250 T ± 12.5 T	nos					-
	7300 T ± 12.5 T	nos					-
	7350 T ± 12.5 T	nos					-
	7400 T ± 12.5 T	nos					-
	7450 T ± 12.5 T	nos					-
	7500 T ± 12.5 T	nos					-
	7550 T ± 12.5 T	nos					-
	7600 T ± 12.5 T	nos					-
	7650 T ± 12.5 T	nos					-
	7700 T ± 12.5 T	nos					-
	7750 T ± 12.5 T	nos					-
	7800 T ± 12.5 T	nos					-
	7850 T ± 12.5 T	nos					-
	7900 T ± 12.5 T	nos					-
	7950 T ± 12.5 T	nos					-
	8000 T ± 12.5 T	nos					-
	8050 T ± 12.5 T	nos					-
	8100 T ± 12.5 T	nos					-
	8150 T ± 12.5 T	nos					-
	8200 T ± 12.5 T	nos					-
	8250 T ± 12.5 T	nos					-
	8300 T ± 12.5 T	nos					-
	8350 T ± 12.5 T	nos					-
	8400 T ± 12.5 T	nos					-
	8450 T ± 12.5 T	nos					-
	8500 T ± 12.5 T	nos					-
	8550 T ± 12.5 T	nos					-
	8600 T ± 12.5 T	nos					-
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	8700 T ± 12.5 T	nos					-
	8750 T ± 12.5 T	nos					-
	8800 T ± 12.5 T	nos					-
	8850 T ± 12.5 T	nos					-
	8900 T ± 12.5 T	nos					-
	8950 T ± 12.5 T	nos					-
	9000 T ± 12.5 T	nos					-
	9050 T ± 12.5 T	nos					-
	9100 T ± 12.5 T	nos					-
	9150 T ± 12.5 T	nos					-
	9200 T ± 12.5 T	nos					-
	9250 T ± 12.5 T	nos					-
	9300 T ± 12.5 T	nos					-
	9350 T ± 12.5 T	nos					-
	9400 T ± 12.5 T	nos					-
	9450 T ± 12.5 T	nos					-
	9500 T ± 12.5 T	nos					-
	9550 T ± 12.5 T	nos					-
	9600 T ± 12.5 T	nos					-
	9650 T ± 12.5 T	nos					-
	9700 T ± 12.5 T	nos					-
	9750 T ± 12.5 T	nos					-
	9800 T ± 12.5 T	nos					-
	9850 T ± 12.5 T	nos					-
	9900 T ± 12						

Project Road : Changtongya Longleng Road

BOQ Item No.

2.01 Project Road Locations not covered under BoQ item 2.02 , 2.03 and 2.04

Cut Quantity	18,01,205	(Summation of Volume under Col. 2 of Annexure 2.1 (b) minus Total Subgrade shape volume under Col. 10)
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2.02 Project Road Locations having Sand Stone away by more than 200 m of habitated area

From	To	Cut Quantity	Remarks
640	1000	9,259	Summation of respective Volumes under Col. 2 of Annexure 2.1 (b)
7225	7350	3,651	
7711	7811	1,586	
26980	27080	1,145	
28980	29270	38,904	
30320	30410	1,568	
30710	30950	17,647	
37870	38370	6,058	
Total		79,817	

2.03 Project Road Locations having Sand Stone within 200 m of habitated area

From	To	Cut Quantity	Remarks
160	250	2,539	Summation of respective Volumes under Col. 2 of Annexure 2.1 (b)
250	420	6,044	
460	640	3,947	
37870	38370	6,058	
Total		18,588	

2.04 Project Road Locations having Sand Stone within 20 m of habitated area

From	To	Cut Quantity	Remarks
0	160	1,536	Summation of respective Volumes under Col. 2 of Annexure 2.1 (b)
420	460	907	
Total		2,442	

2.05 Embankment Construction

1,70,935 (Summation of Volumes under Col. 3 of Annexure 2.1 (b))

2.06 Subgrade Preparation from Borrow Earth (CBR > Designed CBR)

67,839 (30% of Summation of Volumes under Col. 10 of Annexure 2.1(b))

2.07 Preparation of Subgrade in Cut Formation (Loosening and Compacting to full depth)

1,50,956 (70% of Summation of Volumes under Col. 10 of Annexure 2.1(b))

2.08 Preparation of Subgrade in Rocky Formation

14,670 Surface Area for locations having Sandstone Rock

3.01 GSB

78,131 summation of Volumes in Col. 9 in Annexure 2.1 (b)

3.02 WMM

1,05,332 summation of Volumes in Col. 8 in Annexure 2.1 (b)

4.01 Prime Coat

4,12,374 summation of Surface Area in Col. 7 in Annexure 2.1 (b)

4.02 Tack Coat

4,10,827 summation of Surface Area in Col. 5 in Annexure 2.1 (b)

4.03 DBM

20,619 summation of Volumes in Col. 6 in Annexure 2.1 (b)

4.04 BC

12,325 summation of Volumes in Col. 4 in Annexure 2.1 (b)

8.06 Seeding and Mulching

54,457 summation of Cutslope Surface Areas for Category II & III stretches as per slope stability surveys

8.07 Geo-synthetic Mat

13,129 summation of Cutslope Surface Areas under Landslide prone stretches

8.08 Nailing

11,426 summation of Cutslope Surface Areas under Landslide prone stretches

8.09 Temporary Facing

4,050 summation of Surface Areas for first Cut height in Landslide prone stretches

Annexure 2.1 (b) : Template Shape / Component Report (from Mx Roads) : C L Road										Cut Slope				
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	<12 m Sur. Area	> 12 m Sur. Area	RE Wall Sur. Area	R Wall Height
1														
0														
1	15.6	1.8	0.4	13.3	0.6	12.0	3.3	2.5	7.5					
10	65.0	35.2	2.7	90.0	4.6	92.0	23.6	17.7	47.6					2.5
15	10.7	41.5	1.6	53.3	2.7	54.0	13.7	10.1	28.8					3.1
16	2.0	13.1	0.4	13.3	0.7	14.0	3.4	2.5	7.2					3.3
20	4.8	59.8	1.3	43.3	2.2	44.0	11.2	8.2	23.5					4.5
29	3.8	262.3	2.9	96.7	4.9	98.0	24.9	18.3	52.3					6.6
30		51.2	0.4	13.3	0.7	14.0	3.4	2.5	7.2					6.8
30		14.8	0.1	3.3	0.2	4.0	1.0	0.7	2.0					6.9
33		165.4	1.2	40.0	1.9	38.0	9.8	7.2	20.6					7.4
40		391.6	2.3	76.7	3.8	76.0	19.2	14.1	40.4					8.0
47		458.7	2.2	73.3	3.8	76.0	19.1	14.0	40.1					8.3
50		206.7	1.0	33.3	1.7	34.0	8.6	6.3	18.0					8.0
54		245.5	1.4	46.7	2.3	46.0	11.6	8.5	24.4					7.4
60		252.1	1.7	56.7	2.9	58.0	14.6	10.7	30.8					6.6
62		67.4	0.6	20.0	0.9	18.0	4.7	3.5	9.9					6.3
69	3.4	209.2	2.3	76.7	3.8	76.0	19.2	14.1	40.5					5.2
70	0.6	13.2	0.2	6.7	0.3	6.0	1.6	1.2	3.4					5.1
80	44.7	153.1	3.1	103.3	5.2	104.0	26.2	19.2	55.2					3.9
82	14.4	16.5	0.5	16.7	0.9	18.0	4.5	3.3	9.5					3.8
83	11.6	10.5	0.4	13.3	0.6	12.0	3.2	2.3	6.7					3.8
90	90.4	60.0	2.3	76.7	3.9	78.0	19.6	14.4	41.3					3.9
100	162.2	77.0	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.7
101	25.4	10.5	0.5	16.7	0.8	16.0	4.0	2.9	8.4					3.7
110	150.8	65.1	2.8	93.3	4.7	94.0	23.8	17.5	50.0					3.8
120	161.0	87.8	3.2	106.7	5.4	108.0	27.4	20.1	57.7					4.1
120	2.4	1.6	0.1	3.3	0.1	2.0	0.4	0.3	0.9					4.1
130	145.3	97.0	3.0	100.0	5.1	102.0	25.9	19.0	54.5					4.3
140	167.7	90.1	3.0	100.0	5.0	100.0	25.5	18.8	53.9					4.0
143	66.3	26.1	1.0	33.3	1.7	34.0	8.8	6.5	18.6					3.8
150	139.5	43.1	2.0	66.7	3.3	66.0	16.7	12.3	35.3					3.5
160	248.2	51.0	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.1
168	225.4	33.2	2.5	83.3	4.2	84.0	21.1	15.5	44.6					2.9
170	47.1	6.6	0.5	16.7	0.9	18.0	4.4	3.2	9.3					3.0
180	258.3	40.7	2.9	96.7	4.9	98.0	25.1	18.4	52.9					3.1
180														2.9
180	4.6	0.7	0.1	3.3	0.1	2.0	0.5	0.3	1.0					3.1
185	129.4	23.7	1.6	53.3	2.6	52.0	13.2	9.7	27.9					3.2
190	117.6	23.9	1.5	50.0	2.4	48.0	12.3	9.1	26.0					3.3
199	211.7	45.7	2.6	86.7	4.4	88.0	22.1	16.3	46.7					3.4
200	33.1	7.1	0.4	13.3	0.7	14.0	3.4	2.5	7.2					3.4
210	260.0	50.4	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.3
210	7.4	1.3	0.1	3.3	0.1	2.0	0.7	0.5	1.5					3.3
214	93.5	15.5	1.0	33.3	1.7	34.0	8.7	6.4	18.3					3.2
214	2.8	0.4			0.1	2.0	0.3	0.2	0.5					3.2
220	180.9	27.7	1.9	63.3	3.1	62.0	15.9	11.7	33.6					3.3
229	259.4	42.9	2.6	86.7	4.4	88.0	22.4	16.5	47.3					3.5
230	35.9	6.6	0.4	13.3	0.6	12.0	3.1	2.3	6.6					3.5
239	298.8	53.8	2.9	96.7	4.9	98.0	24.9	18.3	52.6					3.7
240	17.9	3.2	0.2	6.7	0.3	6.0	1.4	1.0	3.0					3.7
241	28.0	4.9	0.3	10.0	0.4	8.0	2.2	1.6	4.7					3.7
242	50.3	8.8	0.5	16.7	0.8	16.0	4.0	2.9	8.3					3.7
250	277.1	47.3	2.5	83.3	4.2	84.0	21.3	15.7	44.9					3.6
260	373.2	59.6	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.6
270	394.9	51.2	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.2
272	75.5	7.8	0.6	20.0	1.0	20.0	5.1	3.7	10.6					3.1
280	360.0	29.1	2.7	90.0	4.5	90.0	22.7	16.7	47.8					2.7
290	457.8	32.2	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.0
300	427.8	44.9	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.6
310	383.2	63.3	3.3	110.0	5.5	110.0	27.8	20.4	58.4					4.0
320	364.1	73.9	3.3	110.0	5.5	110.0	27.8	20.4	58.4					4.1
330	351.8	77.3	3.3	110.0	5.5	110.0	27.8	20.4	58.4					4.2
340	348.5	73.0	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.9
350	364.6	64.3	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.8
360	370.9	62.7	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.8
370	352.3	69.2	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.9
380	309.0	69.6	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.7
390	285.8	63.4	3.3	110.0	5.5	110.0	27.8	20.4	58.4					3.6
393	84.5	18.0	1.0	33.3	1.6	32.0	8.1	6.0	17.1					3.6
394	35.3	7.5	0.4	13.3	0.7	14.0	3.4	2.5	7.1					3.6
397	72.0	15.5	0.8	26.7	1.4	28.0	6.9	5.1	14.6					3.6
400	96.0	21.1	1.1	36.7	1.8	36.0	9.3	6.8	19.6					3.7
406	179.4	41.8	2.1	70.0	3.5	70.0	17.8	13.1	37.5					3.7
410	96.9	24.0	1.2	40.0	2.0	40.0	9.9	7.3	20.9					3.7
412	43.6	11.1	0.5	16.7	0.9	18.0	4.6	3.3	9.6					3.7
420	207.0	52.6	2.6	86.7	4.3	86.0	22.0	16.1	46.2					3.6
420	2.5	0.6			0.1	2.0	0.3	0.2	0.6					3.4
420	7.1	1.6	0.1	3.3	0.2	4.0	0.8	0.6	1.7					3.4
421	34.3	7.9	0.5	16.7	0.8	16.0	3.9	2.8	8.1					3.3
430	206.3	45.7	2.7	90.0	4.5	90.0	22.7	16.7	47.8					3.3
435	109.5	23.9	1.4	46.7	2.3	46.0	11.8	8.7	25.0					3.3
440	124.4	28.8	1.6	53.3	2.7	54.0	13.8	10.1	29.1					3.3
450	219.2	53.9	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.2
455	99.5	24.8	1.4	46.7	2.3	46.0	11.9	8.8	25.2					3.3
460	113.4	27.5	1.6	53.3	2.7	54.0	13.6	10.0	28.7					3.2
470	213.3	49.7	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.1
480	214.8	48.1	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.1
490	215.7	48.3	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.2
500	217.0	48.5	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.1
509	206.0	44.7	2.8	93.3	4.7	94.0	24.1	17.7	50.8					3.1
510	12.7	2.7	0.2	6.7	0.3	6.0	1.5	1.1	3.1					3.1
520	219.6	47.7	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.2
530	217.7	49.7	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.2
533	75.2	17.5	1.0	33.3	1.7	34.0	8.8	6.5	18.6					3.2
533														3.0
540	145.5	30.0	2.0	66.7	3.3	66.0	16.7	12.3	35.3					3.2
550	230.7	45.9	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.0
560	223.2	50.1	3.0	100.0	5.0	100.0	25.5	18.8	53.9					3.4
566	115.4	35.1	1.7	56.7	2.9	58.0	14.7	10.8	31.1					3.6
570	79.5	29.0	1.3	43.3	2.1	42.0	10.8	8.0	22.8					3.7
580	188.6	75.4	3.0	100.0	5.0	100.0	25.5	18.8	53.9					4.0
584	70.2	28.2	1.1	36.7	1.8	36.0	9.2	6.8	19.4					4.0
587	70.8	27.7	1.1	36.7	1.8	36.0	9.2	6.7	19.3					3.9
590	56.0	21.3	0.8	26.7	1.4	28.0	7.2	5.3	15.2					3.9
600	210.3	69.9	3.0											

Annexure 2.1 (b) : Template Shape / Component Report (from Mix Roads) : C L Road									
Station	Volume Cut	Volume Fill	Volume BC	Tack Coat	DBM	Surf. Coat	Surf. Area	Subgrade	ESB
21500	495.4	5.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21510	579.8	2.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21520	681.2	0.3	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21530	852.3	0.1	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21540	1004.7	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21550	978.0	1.6	3.1	103.3	5.2	104.0	104.0	27.0	20.4
21560	772.4	3.0	3.1	103.3	5.2	104.0	104.0	27.0	20.4
21570	570.2	8.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21580	459.2	8.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21590	379.5	9.5	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21600	339.6	10.9	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21610	295.9	20.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21620	215.7	45.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21630	134.7	85.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21640	91.1	138.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21650	95.8	165.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21660	120.0	150.3	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21670	150.3	123.3	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21680	197.1	91.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21690	250.8	65.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21700	301.8	49.4	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21800	814.4	9.5	3.3	110.0	5.5	110.0	110.0	27.7	20.4
21803	301.5	1.2	3.3	110.0	5.5	110.0	110.0	27.7	20.4
21810	742.9	7.3	3.6	120.0	5.8	120.0	120.0	28.8	21.5
21820	1145.2	1.9	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21830	1221.5	5.1	3.1	103.3	5.2	104.0	104.0	27.0	20.4
21840	1387.0	2.1	3.1	103.3	5.2	104.0	104.0	27.0	20.4
21850	1554.5	3.1	3.1	103.3	5.2	104.0	104.0	27.0	20.4
21860	1790.3	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21870	2105.3	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21880	2497.6	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21890	2868.4	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21900	3344.6	1.3	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21910	3048.1	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21920	3076.2	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21930	3103.1	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21940	3125.2	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21950	2989.0	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21960	2606.6	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21970	2163.9	3.0	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21975	978.0	1.6	3.0	100.0	5.1	102.0	102.0	26.0	19.5
21980	793.8	1.4	3.1	103.3	5.3	106.0	106.0	28.0	21.5
21990	1415.3	0.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
21991	60.9	0.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22000	1014.2	1.5	3.1	113.3	5.7	114.0	114.0	29.0	21.7
22010	755.8	5.4	3.4	113.3	5.7	114.0	114.0	29.0	21.7
22020	409.3	17.3	3.3	110.0	5.5	110.0	110.0	27.7	20.4
22030	219.3	55.1	3.3	110.0	5.5	110.0	110.0	27.7	20.4
22040	88.8	143.9	3.3	110.0	5.5	110.0	110.0	27.7	20.4
22050	15.9	326.3	3.3	110.0	5.5	110.0	110.0	27.7	20.4
22060	630.3	576.1	3.3	110.0	5.5	110.0	110.0	27.7	20.4
22067	52.1	6.7	6.7	33.3	1.7	34.0	34.0	8.8	6.5
22070	361.6	1.0	3.1	103.3	5.1	102.0	102.0	26.0	19.5
22080	1047.8	3.1	3.1	103.3	5.1	102.0	102.0	26.0	19.5
22082	185.3	0.5	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22090	837.9	2.5	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22100	641.2	3.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22110	326.5	3.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22120	101.9	38.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22130	252.2	13.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22140	305.7	8.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22150	305.7	8.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22160	302.2	8.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22170	315.9	7.3	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22180	330.0	6.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22190	394.4	6.0	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22200	455.1	5.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22210	419.8	4.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22220	370.4	6.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22230	345.6	6.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22240	315.0	7.7	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22250	292.7	10.7	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22260	279.9	13.9	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22270	252.1	17.4	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22280	213.4	24.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22290	183.9	32.7	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22300	165.8	39.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22310	158.4	42.3	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22320	162.3	41.7	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22330	170.2	38.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22340	173.7	37.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22350	174.3	36.9	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22360	174.9	36.7	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22370	175.4	36.4	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22380	168.9	36.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22390	161.0	39.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22400	163.6	39.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22410	171.4	36.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22420	179.5	33.9	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22430	187.7	31.1	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22440	189.3	28.6	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22450	198.0	26.2	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22460	192.9	36.7	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22468	160.6	39.8	3.0	100.0	5.0	100.0	100.0	25.5	18.8
22470	218.1	41.5	3.2	106.7	5.3	106.0	106.0	26.9	19.7
22480	218.1	41.5	3.2	106.7	5.3	106.0	106.0	26.9	19.7
22490	241.5	34.9	3.2	106.7	5.3	106.0	106.0	26.9	19.7

Annexure 2.1 (b) : Template Shape / Component Report (from Mx Roads) : C L Road										Cut Slope		RE Wall		R Wall	
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	<12 m Sur. Area	> 12 m Sur. Area	Sur. Area 14	Sur. Area 15	Height
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
23400	96.0	181.1	2.8	93.3	4.7	110.0	27.8	20.4	50.6	23.0				6.6	7.1
23410	20.6	315.6	3.3	110.0	5.5	110.0	27.8	20.4	58.4	9.0				5.5	4.1
23420	60.3	235.9	3.3	110.0	5.5	110.0	27.8	20.4	58.4	31.0				4.9	5.0
23430	106.2	145.5	3.3	100.0	4.9	98.0	25.1	18.4	52.7	33.0				4.8	3.3
23439	113.9	119.4	3.0	10.0	0.5	10.0	2.6	1.9	5.4	6.0				3.5	3.8
23440	12.6	14.5	0.3	16.7	0.8	16.0	4.3	3.1	47.2	32.0				3.9	4.0
23442	22.7	21.8	0.5	86.7	4.4	88.0	16.9	12.4	35.6	14.0				4.0	4.0
23450	130.9	79.1	2.6	66.7	3.3	66.0	8.8	6.5	18.5	40.0				4.0	4.0
23457	101.9	43.6	2.0	33.3	1.7	34.0	25.5	18.8	53.9	41.0				4.0	4.0
23460	52.0	26.6	1.0	100.0	5.0	100.0	25.5	18.8	53.9	7.0				4.2	4.5
23470	149.0	84.1	3.0	100.0	5.0	100.0	4.6	3.4	9.8	33.0				4.4	4.0
23480	148.9	88.5	3.0	16.7	0.9	18.0	20.9	15.4	44.1	36.0				4.0	4.1
23482	27.1	16.1	0.5	83.3	4.1	82.0	25.5	18.8	53.9	23.0				4.0	4.1
23490	121.7	73.0	2.5	100.0	5.0	100.0	21.4	15.8	45.3	4.0				4.0	4.1
23500	129.7	97.5	3.0	83.3	4.2	84.0	4.1	3.0	8.6	26.0				4.0	4.0
23508	78.4	99.3	2.5	16.7	0.8	16.0	26.2	19.2	55.1	9.0				4.0	4.0
23510	12.2	20.5	0.5	103.3	5.2	104.0	9.2	6.8	19.4	1.0				3.7	2.9
23520	89.3	112.1	3.1	36.7	1.8	36.0	1.4	1.0	2.9	17.0				2.5	2.7
23523	30.2	38.0	1.1	6.7	0.3	6.0	16.8	12.3	35.3	49.0				3.1	3.3
23524	3.7	6.2	0.2	66.7	3.3	66.0	27.7	20.4	58.2	47.0				3.4	3.5
23530	56.1	71.8	3.3	110.0	5.5	110.0	27.7	20.4	58.2	38.0				4.0	3.8
23540	171.5	38.0	3.3	110.0	5.5	110.0	27.7	20.4	58.2	30.0				3.3	3.0
23550	241.2	33.9	3.3	110.0	5.5	110.0	27.7	20.4	58.2	20.0				2.6	3.0
23560	237.3	46.7	3.3	110.0	5.5	110.0	27.7	20.4	58.2	1.0				5.6	5.3
23570	201.9	60.7	3.3	110.0	5.5	110.0	27.7	20.4	58.2	9.0				4.8	4.4
23580	176.6	67.3	3.3	110.0	5.5	110.0	27.7	20.4	58.2	3.0				4.4	4.0
23590	160.8	74.2	3.3	110.0	5.5	110.0	27.7	20.4	58.2	2.0				3.4	3.9
23600	143.1	93.2	3.3	110.0	5.5	110.0	27.7	20.4	58.2	1.0				2.5	2.7
23610	115.0	129.8	3.3	110.0	5.5	110.0	27.7	20.4	58.2	1.0				5.6	5.3
23620	69.1	151.2	3.3	110.0	5.5	110.0	27.7	20.4	58.2	1.0				4.8	4.4
23630	23.4	116.5	3.3	110.0	5.5	110.0	27.7	20.4	58.2	1.0				4.0	4.0
23640	5.9	76.4	3.3	110.0	5.5	110.0	7.9	5.8	16.6	2.9				3.9	3.9
23650	6.1	33.3	0.9	30.0	1.6	32.0	1.4	1.0	37.4	1.0				3.4	2.5
23653	0.6	8.5	0.2	6.7	0.3	6.0	17.8	13.1	45.8	9.0				1.9	1.5
23653							21.7	16.0	8.8	3.0				1.2	1.0
23660	3.8	134.2	2.6	86.7	4.3	86.0	4.2	3.1	53.9	25.0				1.9	1.5
23668	27.3	170.6	0.5	16.7	0.8	16.0	25.5	18.8	53.9	33.0				1.9	1.5
23670	9.3	28.8	3.0	100.0	5.0	100.0	25.5	18.8	53.9	37.0				1.7	1.6
23680	74.3	156.4	3.0	100.0	5.0	100.0	25.5	18.8	53.9	37.0				1.4	1.2
23690	105.3	124.7	3.0	100.0	5.0	100.0	25.5	18.8	53.9	34.0				1.4	1.2
23700	130.9	98.3	3.0	100.0	5.0	100.0	25.5	18.8	53.9	27.0				1.2	1.1
23710	137.4	88.4	3.0	100.0	5.0	100.0	19.5	14.3	41.1	10.0				1.1	1.0
23720	129.6	91.7	2.3	76.7	3.8	76.0				51.0				1.0	0.8
23728	98.5	70.8					24.0	6.1	4.5	12.8				0.5	0.4
23728							24.0	6.1	4.5	12.8				0.5	0.4
23730	34.1	21.6	0.7	23.3	1.2	24.0	25.5	18.8	53.9	76.0				1.9	1.5
23740	199.9	72.1	3.0	100.0	5.0	100.0	25.5	18.8	53.9	159.0				1.2	1.0
23750	302.7	39.7	3.0	100.0	5.0	100.0	25.5	18.8	53.9	175.0				1.0	0.7
23760	389.4	19.5	3.0	100.0	5.0	100.0	25.5	18.8	53.9	146.0				0.5	0.5
23770	450.1	11.4	3.0	100.0	5.0	100.0	25.5	18.8	53.9	46.0				0.5	0.5
23780	515.7	7.4	3.0	43.3	2.1	42.0	10.8	7.9	31.1	22.8				0.5	0.5
23784	240.3	2.4	1.3	56.7	2.9	58.0	14.8	10.9	51.0	26.0				0.4	0.8
23784	383.5	2.5	1.7	96.7	4.8	96.0	2.1	1.5	4.4	127.0				1.6	2.2
23790	719.9	2.6	2.9	6.7	0.4	8.0	10.2	7.5	21.4	217.0				1.6	2.2
23799	63.9	0.2	0.2	40.0	2.0	40.0	17.4	12.7	36.4	346.0				2.1	2.1
23800	312.7	0.7	1.2	66.7	3.4	68.0	27.7	20.3	58.2	268.0				2.1	2.1
23804	541.5	1.1	2.0	110.0	5.5	110.0	27.7	20.3	58.2	146.0				2.1	2.1
23810	841.9	2.6	3.3	110.0	5.5	110.0	23.1	16.9	48.4	5.0				1.9	1.9
23820	621.6	6.5	3.3	90.0	4.6	92.0	4.5	3.3	9.5	18.0				1.9	1.9
23830	373.6	13.3	2.7	16.7	0.9	18.0	7.7	5.7	16.2	39.0				1.9	1.9
23838	55.6	3.6	0.5	30.0	1.5	30.0	18.9	13.9	39.7	43.0				1.7	1.6
23840	84.1	5.8	0.9	73.3	3.7	74.0	20.2	14.8	42.6	12.0				1.5	1.5
23843	200.5	12.4	2.2	80.0	4.0	80.0	5.5	4.0	11.6	57.0				1.4	1.4
23850	230.0	11.9	2.4	20.0	1.1	22.0	25.5	18.8	53.9	59.0				1.2	1.2
23858	65.5	3.1	0.6	100.0	5.0	100.0	25.5	18.8	53.9	61.0				1.4	1.4
23860	315.0	13.0	3.0	100.0	5.0	100.0	25.5	18.8	53.9	27.0				1.2	1.2
23870	332.1	11.0	3.0	100.0	5.0	100.0	11.2	8.3	23.7	35.0				1.1	1.1
23880	349.6	9.3	3.0	43.3	2.2	44.0				64.0				1.0	1.0
23890	159.4	3.7	1.3				56.0	14.3	10.5	30.2				1.0	1.0
23894							25.5	18.8	53.9	31.0				0.8	0.8
23900	209.8	3.8	1.7	56.7	2.8	56.0	25.5	18.8	53.9	56.0				0.8	0.8
23910	386.0	7.0	3.0	100.0	5.0	100.0	25.5	18.8	53.9	50.0				0.8	0.8
23920	413.4	6.2	3.0	100.0	5.0	100.0	2.4	1.8	5.1	70.0				1.0	1.0
23930	428.5	5.5	3.0	10.0	0.5	10.0	23.4	17.2	49.2	8.0				1.4	1.4
23931	40.1	4.1	2.8	93.3	4.6	92.0	15.7	11.6	33.1	138.0				1.4	1.4
23940	376.1	2.3	1.9	63.3	2.2	64.0	11.0	8.1	23.2	110.0				1.5	1.5
23946	256.6	1.5	1.3	43.3	0.2	4.0	1.2	0.9	2.6	95.0				1.7	1.7
23950	193.8	0.2	0.1	3.3	5.2	104.0	26.4	19.4	55.4	18.0				1.4	1.4
23950	22.2	4.4	3.1	103.3	5.2	110.0	27.7	20.3	58.2	62.0				1.8	1.8
23960	526.9	6.4	3.3	110.0	5.5	96.0	24.3	17.8	7.0	122.0				1.9	1.9
23970	624.6	6.4	2.9	96.7	4.8	14.0	3.4	2.5	18.7	144.0				2.1	2.1
23979	545.9	0.9	0.4	13.3	0.7	36.0	8.9	6.5	37.4	29.0				2.3	2.3
23980	71.6	2.5	1.1	36.7	1.8	70.0	17.8	13.0	44.9	165.0				2.4	2.4
23983	185.0	6.1	2.1	70.0	3.5	84.0	21.3	15.7	9.3	65.0				2.6	2.6
23990	342.8	6.1	2.5	83.3	4.2	18.0	4.4	3.2	53.9	23.0				2.7	2.7
23998	373.9	10.0	0.5	16.7	0.9	100.0	25.5	18.8	53.9	37.0				3.0	3.0
24000	72.8	2.5	3.0	100.0	5.0	100.0	25.5	18.8	53.9	61.0				3.1	3.1
24010	401.3	16.3	3.0	100.0	5.0	100.0	25.5	18.8	53.9	64.0				3.2	3.2
24020	357.7	19.4	3.0	100.0	5.0	100.0	25.5	18.8	53.9	77.0				3.3	3.3
24030	322.6	23.0	3.0	100.0	5.0	100.0	9.7	7.1	20.4	155.0				4.3	4.3
24040	303.2	26.9	3.0	100.0	5.0	100.0				1					

Annexure 2.1 (b) : Template Shape / Component Report (from Mx Roads) : C.L Road														
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	Cut Slope <12 m Sur. Area	> 12 m Sur. Area	RE Wall Sur. Area	R Wall Height
1														
26980														
26990	94.4	10.3	3.4	113.3	5.7	114.0	29.0	21.7	58.3					1.0
27000	35.7	26.5	3.3	110.0	5.5	110.0	27.8	20.4	58.3					1.5
27002	4.5	5.6	0.6	20.0	1.0	20.0	4.9	3.6	10.2					1.6
27010	24.2	20.3	2.7	90.0	4.5	90.0	22.9	16.8	48.1					1.3
27019	109.7	9.5	3.0	100.0	5.1	102.0	25.9	19.4	56.8					
27020	23.2	0.8	0.4	13.3	0.6	12.0	3.2	2.4	7.3					
27030	246.3	8.3	3.5	116.7	5.8	116.0	30.2	22.9	68.4					
27040	219.8	7.4	3.5	116.7	5.8	116.0	30.2	22.9	68.4					
27050	104.6	14.3	3.4	113.3	5.7	114.0	29.0	21.6	58.5					1.4
27060	24.8	24.8	3.3	110.0	5.5	110.0	27.8	20.4	58.5					1.5
27070	22.0	24.6	3.3	110.0	5.5	110.0	27.8	20.4	58.5					1.6
27070	0.2	0.1					0.2	0.1	0.4					1.6
27078	26.7	12.5	2.4	80.0	4.0	80.0	20.4	15.0	43.0					1.2
27080	10.7	3.3	0.8	26.7	1.3	26.0	6.5	4.7	13.6					1.1
27080														1.1
27090	48.4	10.3	3.1	103.3	5.1	102.0	25.9	19.1	54.7					0.8
27093	13.0	1.9	0.8	26.7	1.3	26.0	6.5	4.8	13.8					0.8
27099	33.8	4.2	2.0	66.7	3.3	66.0	16.8	12.4	35.6					0.7
27100	4.3	0.5	0.3	10.0	0.4	8.0	2.1	1.6	4.5					0.7
27110	63.3	3.5	3.1	103.3	5.2	104.0	26.8	20.0	59.0					
27114	28.9	0.4	1.2	40.0	2.1	42.0	10.8	8.2	24.7					
27114														
27120	49.4	0.1	2.0	66.7	3.3	66.0	17.2	13.1	39.6	2.0				
27130	100.9		3.2	106.7	5.4	108.0	28.0	21.3	64.1	24.0				
27132	24.5		0.6	20.0	1.0	20.0	5.1	3.9	11.7	8.0				
27140	99.6		2.6	86.7	4.1	88.0	23.0	17.5	52.6	26.0				
27147	63.1	1.2	2.2	73.3	3.8	76.0	19.6	14.9	44.9	10.0				
27150	31.4	3.8	1.1	36.7	1.8	36.0	9.4	7.1	21.4	5.0				
27159	141.6	11.8	3.1	103.3	5.3	106.0	27.1	20.3	54.3	28.0				1.3
27160	13.1	0.4	0.2	6.7	0.4	8.0	1.9	1.4	4.1	3.0				1.4
27170	222.0	7.6	3.5	116.7	5.8	116.0	29.8	22.3	65.5	45.0				
27180	237.2	5.1	3.6	120.0	6.0	120.0	31.2	23.7	71.4		74.0			
27190	387.2	0.4	3.6	120.0	6.0	120.0	31.2	23.7	71.4		151.0			
27200	594.4	0.4	3.6	120.0	6.0	120.0	31.2	23.7	71.4		129.0			
27210	468.1	1.5	3.4	113.3	5.8	116.0	29.6	22.2	59.2		131.0			0.4
27210	11.4	0.1	0.1	3.3	0.2	4.0	1.1	0.8	2.2		6.0			0.4
27210	3.2				0.1	2.0	0.3	0.2	0.6		2.0			0.6
27210														0.9
27220	362.3	5.5	3.1	103.3	5.3	106.0	26.6	19.5	56.0		136.0			0.9
27225	259.9	3.1	1.8	60.0	3.0	60.0	15.2	11.2	32.0		57.0			0.9
27230	218.6	2.2	1.5	50.0	2.5	50.0	13.1	9.7	28.6		64.0			0.9
27230	0.8						0.1		0.1					
27240	383.4	2.2	3.5	116.7	5.8	116.0	30.2	22.9	68.5		130.0			
27250	565.5	0.1	3.5	116.7	5.8	116.0	30.2	22.9	68.5		188.0			
27260	801.7	0.8	3.5	116.7	5.8	116.0	30.2	22.9	68.5		299.0			
27265	301.6	1.1	1.7	56.7	2.9	58.0	14.8	11.2	33.6		100.0			
27270	270.8	2.0	1.7	56.7	2.9	58.0	14.8	11.0	29.8		108.0			0.9
27277	519.7	3.8	2.4	80.0	4.0	80.0	20.2	14.8	42.4		242.0			0.9
27280	237.2	1.4	0.9	30.0	1.5	30.0	7.6	5.6	16.0		96.0			0.8
27290	1109.0	4.7	3.3	110.0	5.5	110.0	27.8	20.4	58.4		280.0			0.7
27300	1627.8	3.7	3.3	110.0	5.5	110.0	27.8	20.4	58.2			275.0		0.5
27306	1383.0	1.6	2.1	70.0	3.5	70.0	17.8	13.0	37.2			394.0		0.3
27310	863.5	0.7	1.2	40.0	2.0	40.0	10.0	7.3	20.9			249.0		0.3
27320	2338.2	3.5	3.3	110.0	5.5	110.0	27.7	20.3	58.1			545.0		1.1
27330	1989.7	5.1	3.3	110.0	5.5	110.0	27.7	20.3	58.0			407.0		1.0
27330	4.0						0.1		0.1			1.0		1.0
27340	1914.2	4.3	3.3	110.0	5.5	110.0	27.7	20.3	58.2			583.0		0.7
27350	2026.4	4.1	3.3	110.0	5.5	110.0	28.0	20.5	58.7			700.0		0.8
27360	1930.1	5.9	3.4	113.3	5.8	116.0	29.6	22.0	64.5			479.0		
27368	1521.0	10.0	2.9	96.7	4.9	98.0	25.1	19.0	56.7			234.0		
27370	337.5	3.3	0.7	23.3	1.1	22.0	5.8	4.4	13.1			53.0		
27380	1567.1	18.0	3.6	120.0	6.0	120.0	31.0	23.4	69.8			394.0		
27390	997.0	46.7	3.5	116.7	5.8	116.0	30.0	22.5	61.9		100.0			
27400	353.1	44.6	3.5	116.7	5.8	116.0	30.0	22.5	66.7	19.0				
27401	6.1	0.9	0.2	6.7	0.4	8.0	2.0	1.5	4.4	2.0				
27410	71.4	10.0	3.3	110.0	5.5	110.0	28.4	21.5	64.1	27.0				
27414	24.0	3.4	1.4	46.7	2.3	46.0	12.1	9.2	27.6	8.0				
27415	5.5	1.0	0.3	10.0	0.6	12.0	2.9	2.2	6.6	2.0				
27420	25.4	6.4	1.6	53.3	2.7	54.0	13.8	10.5	31.7	8.0				
27429	49.3	10.2	2.9	96.7	4.9	98.0	25.7	19.5	58.8	11.0				
27430	5.0	0.6	0.3	10.0	0.5	10.0	2.4	1.8	5.6	1.0				
27430	0.9	0.1			0.1	2.0	0.4	0.3	0.9					
27435	26.9	2.7	1.4	46.7	2.4	48.0	12.6	9.6	28.9	6.0				
27440	34.1	1.7	1.7	56.7	2.9	58.0	15.0	11.4	34.4	9.0				
27450	90.4	0.7	3.2	106.7	5.4	108.0	28.0	21.3	64.1	25.0				
27460	181.5		3.2	106.7	5.4	108.0	28.0	21.3	64.1		39.0			
27464	105.4		1.2	40.0	2.0	40.0	10.2	7.8	23.4		30.0			
27470	505.2		2.0	66.7	3.4	68.0	17.8	13.5	40.7			216.0		
27480	1336.2	11.4	3.1	103.3	5.2	104.0	27.0	20.4	55.8			689.0		
27490	1320.3	11.4	3.1	103.3	5.2	104.0	27.0	20.4	60.9			705.0		
27500	1074.6	0.1	3.2	106.7	5.4	108.0	28.0	21.3	64.1			711.0		
27504	357.5		1.3	43.3	2.2	44.0	11.7	8.9	26.7			295.0		
27510	298.7		1.9	63.3	3.1	62.0	16.4	12.5	37.6		24.0			
27514	81.7		1.3	43.3	2.2	44.0	11.3	8.6	25.8	13.0				
27518	75.5	1.0	1.4	46.7	2.4	48.0	12.6	9.5	26.0	24.0				
27519	10.3	0.3	0.2	6.7	0.4	8.0	1.9	1.4	4.2	3.0				
27520	12.0	0.3	0.3	10.0	0.4	8.0	2.3	1.7	5.1	3.0				
27524	50.2	0.5	1.2	40.0	2.0	40.0	10.2	7.6	22.5	10.0				
27530	83.6		2.1	70.0	3.5	70.0	17.9	13.4	39.3	6.0				
27533	45.2		1.1	36.7	1.9	38.0	9.8	7.3	21.5					
27540	81.7		2.1	70.0	3.6	72.0	18.5	13.8	40.6	1.0				
27543	40.0		1.1	36.7	1.9	38.0	9.6	7.1	21.0					
27548	51.1	0.4	1.5	50.0	2.5	50.0	13.0	9.8	29.1					
27550	22.8	0.6	0.7	23.3	1.2	24.0	6.2	4.7	14.2					
27560	112.2	2.1	3.3	110.0	5.5	110.0	28.6	21.8	65.5	2.0				
27563	35.4	0.1	0.9	30.0	1.6	32.0	8.1	6.2	18.6	1.0				
27570	94.9	0.1	2.3	76.7	3.8	76.0	19.9	15.1	45.6	6.0				
27580	161.6		3.2	106.7	5.4	108.0	28.0	21.3	64.1	17.0				
27590	256.0		3.0	100.0	5.0	100.0	25.9	19.5	53.4	37.0				
27590	15.2		0.1	3.3	0.2	4.0	1.1	0.8	2.4	2.0				
27600	445.5		3.0	100.0	5.1	102.0	26.0	19.5	57.6		94.0			
27601	66.6		0.4	13.3	0.7	14.0	3.4	2.5	7.5					

QUANTITY CALCULATION FOR 1X1.2 DIA PIPES

Sl.No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc Foundation	Conc Substructure	PCC M15 Levelling coarse	Filter Media	Painting	Lettering
1	25.435	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
2	4.570	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
3	8.270	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
4	37.420	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
5	31.948	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
6	9.620	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
7	8.320	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
8	31.872	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
9	9.560	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
10	36.330	1*1.2	12.50	148.02	179.18	14.58	24.00	3.88	33.84	2.28	50.00
11	9.690	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
12	15.960	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
13	7.640	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
14	31.739	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
15	5.980	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
16	7.600	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
17	8.640	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
18	34.220	1*1.2	12.50	148.02	179.18	14.58	24.00	3.88	33.84	2.28	50.00
19	12.870	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
20	6.120	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
21	8.760	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
22	35.270	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
23	35.750	1*1.2	12.50	148.02	179.18	14.58	24.00	3.88	33.84	2.28	50.00
24	25.350	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
25	21.200	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
26	9.860	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
27	21.530	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
28	7.380	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00
29	27.548	1*1.2	10.00	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50.00

QUANTITY CALCULATION FOR 1X1.2 DIA PIPES

Sl.No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc Foundation	Conc Substructure	PCC M15 Levelling coarse	Filter Media	Painting	Lettering
142	0.740	1*1.2	10.00	159.87	216.76	17.15	27.93	4.19	37.44	2.28	50.00
143	24.215	1*1.2	10.00	159.87	216.76	17.15	27.93	4.19	37.44	2.28	50.00
144	1.745	1*1.2	10.00	159.87	216.76	17.15	27.93	4.19	37.44	2.28	50.00
145	1.630	1*1.2	12.50	159.87	216.76	17.15	27.93	4.19	37.44	2.28	50.00
146	1.330	1*1.2	10.00	178.81	276.06	21.03	33.42	4.68	41.04	2.28	50.00
147	31.400	1*1.2	12.50	70.22	276.06	21.03	33.42	4.68	41.04	2.28	50.00
148	14.870	1*1.2	10.00	178.81	276.06	21.03	33.42	4.68	41.04	2.28	50.00
149	1.770	1*1.2	12.50	178.81	276.06	21.03	33.42	4.68	41.04	2.28	50.00
150	17.770	1*1.2	12.50	178.81	276.06	21.03	33.42	4.68	41.04	2.28	50.00
151	33.990	1*1.2	10.00	178.81	276.06	21.03	33.42	4.68	41.04	2.28	50.00
152	1.190	1*1.2	12.50	178.81	276.06	21.03	33.42	4.68	41.04	2.28	50.00
153	13.300	1*1.2	10.00	204.87	349.53	27.46	39.54	5.36	44.64	2.28	50.00
154	0.065	1*1.2	10.00	177.33	349.53	27.46	39.54	5.36	44.64	2.28	50.00
155	11.700	1*1.2	12.50	204.87	349.53	27.46	39.54	5.36	44.64	2.28	50.00
156	11.300	1*1.2	10.00	204.87	349.53	27.46	39.54	5.36	44.64	2.28	50.00
		Total	1747.50	21784.31	26226.82	2278.80	3624.42	585.25	5117.04	355.68	7800.00

QUANTITY CALCULATION FOR 2X1.2 DIA PIPES

Sl.No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc Foundation	Conc Substructure	PCC M15 Levelling coarse	Filter Media	Painting	Lettering
157	5.310	2*1.2	25.000	132.630	137.214	18.15	21.74279674	3.596	30.24	2.28	50
158	28.434	2*1.2	20.000	121.694	137.214	18.15	21.74279674	3.596	30.24	2.28	50
159	9.390	2*1.2	25.000	132.630	137.214	18.15	21.74279674	3.596	30.24	2.28	50
160	30.060	2*1.2	25.000	77.946	137.214	18.15	21.74279674	3.596	30.24	2.28	50
161	15.350	2*1.2	25.000	132.630	137.214	18.15	21.74279674	3.596	30.24	2.28	50
162	32.320	2*1.2	20.000	132.630	137.214	18.15	21.74279674	3.596	30.24	2.28	50
163	30.570	2*1.2	25.000	76.210	137.214	18.15	21.74279674	3.596	30.24	2.28	50
164	32.490	2*1.2	25.000	132.630	137.214	18.15	21.74279674	3.596	30.24	2.28	50

QUANTITY CALCULATION FOR 2X.0X2.0X2.0 BOX CULVERTS

S. No.	Proposed Chainage	Designation	Excavation	RCC for Box	Approach slab rcc	Exp. Gap	Lettering	Drainage Spout	Painting	Levelling course	Excavation for Return Wall	RCC for Return Wall	Filter Media	Backfill	Rcc For Counterfort	Rcc For Wall
	(km)		m ³	m ³	m ³	lm	nos.	nos.	m ²	m ³	m ³	m ³	m ³	m ³	m ³	m ³
1	0.622	1.22.00	56.05	52.19	24.50	20.00	50.00	4.00	9.88	14.59	69.55	29.2	24.96	147.4309	0.55	3.93
2	2.370	1.22.00	56.05	52.19	24.50	20.00	50.00	4.00	9.88	14.59	55.64	22.24	20.16	91.47578	0.13	0.91
3	8.050	1.22.00	114.01	59.39	28.18	23.00	50.00	4.00	9.88	16.73	79.23	22.24	20.16	91.47578	0.00	0.00
4	17.120	1.22.00	59.41	55.07	25.97	21.20	50.00	4.00	9.88	15.44	69.55	29.2	24.96	147.4309	0.44	3.20
5	21.730	1.22.00	72.06	55.07	25.97	21.20	50.00	4.00	9.88	15.44	62.17	22.24	20.16	91.47578	0.00	0.00
6	21.910	1.22.00	100.70	52.19	24.50	20.00	50.00	4.00	9.88	14.59	80.08	22.24	20.16	91.47578	0.00	0.00
7	22.070	1.22.00	11.40	52.19	24.50	20.00	50.00	4.00	9.88	14.59	98.40	105.4	48.96	675.1056	3.48	25.05
8	23.570	1.22.00	59.41	55.07	25.97	21.20	50.00	4.00	9.88	15.44	69.55	29.2	24.96	147.4309	0.35	2.54
Total			529.10	433.37	204.09	166.60	400.00	32.00	79.04	121.41	584.17	281.96	204.48	1483.30	4.95	35.63

QUANTITIES FOR PROTECTION WORK FOR 1x1.2 DIA PIPES

SL. No.	DOWNSTREAM PROTECTION WORKS										UPSTREAM PROTECTION WORKS									
	Proposed Chainage	Excavation (km)	PCC M15	RCC for flooring+ wall +block	Rcc in Front and Back Curtain Wall	RCC Total	Hand woven Wire Crates	Reinforce ment	Excavation Catchpit	Excavation Total	PCC M15	RCC	RCC For Catchpit	RCC Total	Hand Woven Wire Crates	Reinforce ment	Reinforce ment for Catchpit	Reinforce ment Total		
1	25.435	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
2	4.570	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
3	8.270	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
4	37.420	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
5	31.948	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
6	9.620	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
7	8.320	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
8	31.872	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
9	9.560	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
10	36.330	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
11	9.690	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
12	15.960	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
13	7.640	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
14	31.739	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
15	5.980	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
16	7.600	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
17	8.640	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
18	34.220	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
19	12.870	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
20	6.120	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
21	8.760	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
22	35.270	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
23	35.750	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
24	25.350	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
25	21.200	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
26	9.860	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
27	21.530	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
28	7.380	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
29	27.548	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
30	8.840	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
31	24.570	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
32	7.350	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
33	20.150	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
34	28.925	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
35	30.730	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	

**Detailed Calculation For Upstream Protection Works for 2mx2m Single cell Box
Culvert**

SL NO.	ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD
		Segment 1									
1)	5.01	Earthwork in excavations:									
		For RCC Rigid flooring (M-30)		m ³	1.00	3.00	4.40	13.20	0.60	7.92	7.92
		For Hand woven wire crates in flexible apron for protection work in upstream		m ³	1.00	3.00	5.40	16.20	1.30	21.06	21.06
		For curtain wall in segment 1		m ³	1.00	5.40	1.70		1.40	12.85	12.85
										Sub Total	41.83
		70% in Ordinary Soil									29.28
		30% in Hard Rock									12.55
2	5.02	PCC M-15 below curtain wall	100 mm thick.	m ³	1.00	5.40	1.70		0.10	0.92	0.92
3	5.02	PCC M-15 below RCC flooring	100 mm thick.	m ³	1.00	3.00	4.40	13.20	0.10	1.32	1.32
										Sub Total	2.24
4	5.04	RCC rigid flooring M-30			1.00	3.00	4.40	13.20	0.20	2.64	2.64
5	5.04	RCC Side walls over rigid flooring M-30	0.2 m thick.	m ³	2.00	3.16	0.20		1.00	1.26	1.26
										Sub Total	3.90
6	5.04	Rcc Curtain wall 1		m ³	1.00	5.40		0.84		4.54	4.54
7	5.20	Hand woven wire crates in flexible apron for protection work	1000 mm thick.	m ³	1.00	3.00	5.40	16.20	1.00	16.20	16.20
8	5.07	Reinforcement in Rcc curtain wall		t	1.00	4.54	@50kg/m3 of concrete			226.80	0.227
9	5.07	Reinforcement in RCC flooring		t	1.00	3.90	@50kg/m3 of concrete			195.25	0.195
		Segment-2									
1)	5.01	Earthwork in excavations:									
		Rcc slope protection structure in segment 2		m ³	1.00	1.00	3.40		0.60	2.04	2.04
		For curtain wall in segment 2		m ³	1.00	3.40	0.40		1.50	2.04	2.04
										Sub Total	2.04
		50% in Ordinary Soil									1.02
		50% in Hard rock									1.02
2	5.02	Pcc below Curtain wall			1.00	3.40	0.40		0.10	0.14	0.14
3	5.02	Pcc below Concrete for slope protection structure	(per meter)	m ³	1.00	1.00	3.40		0.10	0.34	0.34
										Sub Total	0.48
4	5.04	Rcc Concrete for slope protection structure	(per meter)	m ³	1.00	1.00	3.40		0.20	0.68	0.68
5		RCC Side walls over rigid flooring M-30	(per meter)	m ³	2.00	1.00	0.20		1.00	0.40	0.40
6	5.04	Rcc Curtain wall		m ³	1.00	3.40	0.3		1.50	1.53	1.53
7	5.04	Buffle pier		m ³	28.00	0.20	0.20		0.20	0.224	0.224
										Sub Total	2.83
8	5.07	Reinforcement in Rcc curtain wall		t	1.00	1.5300	@50kg/m3 of concrete			76.50	0.077
9	5.07	Reinforcement in RCC slope protection structure		t	1.00	1.30	@50kg/m3 of concrete			65.20	0.065
		Segment 3									
1)	5.01	Earthwork in excavations:									
		For catchpit		m ³	1.00	1.95	3.80	7.41	1.30	9.63	9.63
		50% in Ordinary Soil									4.82
		50% in Hard Rock									4.82
2	5.02	PCC M-15 below Catchpit		m ³	1.00	1.85	3.70	6.85	0.10	0.68	0.68
3	5.04	Rcc M-30 Catchpit walls	A1 (per meter)	m ³	2.00	1.50	0.20		1.00	0.60	0.60
			A2 (per meter)	m ³	1.00	3.40	0.20		1.00	0.68	0.68
										Sub Total	1.28
4	5.04	RCC M-30 Catchpit base		m ³	1.00	1.85	3.90		0.20	1.44	1.44
5	5.07	Reinforcement in Rcc catchpit walls		t	1.00	1.28	@50kg/m3 of concrete			64.00	0.064
6		Reinforcement in Rcc catchpit base		t	1.00	1.44	@50kg/m3 of concrete			72.15	0.072

Detailed Calculation For Downstream Protection Works for 3mx3m Single cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD TOTAL
5.1	Earthwork in excavations for protection works									
	For RCC Rigid flooring (M-30)		m ³	1.00	5.00	5.40	27.00	0.30	8.10	8.10
	For Hand woven wire crates in flexible apron for protection work		m ³	1.00	6.00	6.40	38.40	1.00	38.40	38.40
	For front curtain wall		m ³	1.00	6.40	1.70		1.30	14.14	14.14
	For back curtain wall		m ³	0.00	4.40	0.30		1.00	0.00	0.00
										60.64
	70% in soil									42.45
	30% in ordinary rock									18.19
	Outlet/downstream Protection Work									
5.04	RCC channel/flooring M-30	0.2m thick	m ³	1.00	5.00	5.40		0.20	5.40	5.40
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m ³	36.00	0.30	0.20		0.30	0.65	0.65
		Sub Total								8.09
5.04	RCC in Front Curtain wall		m ³	1.00	6.40		0.81		5.18	5.18
5.04	RCC in Back Curtain wall		m ³	1.00	4.40	0.30		1.00	1.32	1.32
		Sub Total								6.50
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m ³	1.00	5.00	5.40		0.10	2.70	2.70
5.02	PCC M-15 below front Curtain wall		m ³	1.00	6.40	1.70		0.10	1.09	1.09
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m ³	1.00	6.00	6.40		1.00	38.40	38.40
5.07	Reinforcement in RCC flooring		MT	1.00	8.09		@50kg/m ³ of concrete		404.38	0.404
5.07	Reinforcement in curtain wall		MT	1.00	6.50		@50kg/m ³ of concrete		325.20	0.325

Summary Of Quantities for bridges of CL road

Item No. as per Quantity Estimates	Item No. as per BOQ	Description	Unit	RCC voided Slab at KM. 6.580	RCC voided Slab at KM. 9.230	RCC Slab at KM 10.418	Total quantity	Item No. in Rate Analysis	Remarks
1	2	3	4	5	6	7	8	9	10
6.01	6.01	BILL NO. 6: NEW BRIDGES Excavation for Foundation of Structure, as per drawings and Technical Specification, in							
	a	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	643	1072	1537	3251	(3.32+3.33)/2	
	b	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum					3.34	
	c	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum					3.9	
	d	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	780	459	659	1898	3.8 A	
6.02	6.02	PCC (M 15 Grade) in Levelling course below Foundation, Pile Cap, Approach slab etc. complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	67	59	66	191	14.1C	
6.03	6.03	RCC / PCC in open foundations excluding reinforcement complete as per drawings	cum						
	a	M 30 Grade	cum	468	363	444	1275	12.8 G case - I	RCC rates are taken
	b	M 35 Grade	cum					12.8 H case - I	RCC rates are taken
6.04	6.04	RCC / PCC in Substructure, Wing Wall, retaining wall, Box Cell Structures, Seismic restrainer etc. excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200							
	a	M 25	cum					13.5 F (p) Case-I	
	b	M 30	cum					13.5 G (p) Case-I	
	c	M 35	cum	402	395	562	1360	13.5 F (h) Case-I	
6.05	6.05	RCC in Superstructure excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2300.							
	a	Solid Slab (M 30 Grade)	cum			80	80	14.1C Case I (i) q	
	b	Deck Slab for Composite Girder (M35 Grade)	cum					14.1D Case I (ii) q	
	c	Voided Slab (M35 Grade)	cum	303	160		463	14.1D Case I (i) q	
6.06	6.06	Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600.	cum	167	138	116	421	(12.40+13.6+14.2)/3	Average of three
6.07	6.07	Steel Girder for Steel Composite Superstructure including railing and fixing of girder with Bearing complete as per drawings and Technical Specification 1000 and 1900.	MT						
			MT					14.25(i)	
6.08	6.08	Bituminous (Type 2) Wearing Coat as per drawings and Technical Specification Section 2700.	sqm	315	179	128	621	((5.8*0.04) for grading I)+5.14)	
6.09	6.09	40 thk. PCC (M25) finished with 15 thk plaster (1:3) complete as per drawings and Technical Specification.	cum	5	3	2	10	13.5 F (p) Case-I	
6.10	6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000							
	a	Tar Paper Bearings	sqm						
	b	Elastomeric Bearings	cum			37	37	As per Market	
	c	POT cum PTEE Bearings						13.14	
	i	Free / Guided Bearings, having Capacity of							
		575 T ± 25 T	nos					As per Market	
		525 T ± 25 T	nos					As per Market	
		325 T ± 25 T	nos					As per Market	
		275 T ± 25 T	nos	4	4		8	As per Market	

QUANTITY CALCULATION FOR BRIDGE AT KM - 6.580 (C-L Road)			Total Quantity	Unit
No. of Abutments = 2 Abutment				
1. Earthwork in excavation for foundation for structures in all types of soil excluding rock:				
Abutment	=	337.680 m ³	675.36	m ³
2. Filling loose soil in foundations & plinth, including consolidation by watering & ramming in 15cm layers				
Abutment	=	0.000 m ³	0	m ³
Backfilling behind abutments:	=	386.782 m ³	773.56	m ³
3. Structural cement concrete for Plain/ Reinforced Concrete in pile cap/open foundation: R.C.C M35 :				
Abutment 7 x 13.2 x 1	=	92.4 m ³	184.8	m ³
4. Structural cement concrete for Plain/ Reinforced Concrete in substructure, pier, piercap, dirt wall, pedestals, abutment, abutmentcap, retaining wall, return wall, staircase, etc.	Total :	83.999 m ³	168.00	m ³
RCC M35 in abutment:-				
Dirt Wall	=	3.185 m ³		
Abutment Cap	=	8.554 m ³		
Abutment	=	47.840 m ³		
Return wall	=	24.420 m ³		
5. Plain cement concrete M 15 leveling course below foundation below approach slabs for bridges :				
Below foundation : 7.2 x 13.4 x 0.15	=	9.648 m ³		
Below approach : 3.5 x 0.15 x 13	=	6.825 m ³		
	Total :	16.473 m ³	32.946	m ³
6. Structural concrete grade M30 in Reinforced Cement Concrete in approach slabs excluding reinforcement :				
3.5 x 13 x 0.35	=	15.93 m ³	31.85	m ³
7. Providing & fixing in position HYSD reinforcement bars				
a)- Foundation/pile cap, pile, dirt wall & restrainer	=	11.805 T		
b)- Abutment and Return Wall	=	7.228 T		
c)- Abutment Cap	=	5.132 T		
d)- Approach slab	=	0.000 T		
	TOTAL =	24.163 T	48.33	T
	Say :	24.16 T	0	cm ³
8. Elastomeric Stopper Pad =				
9. Weep Holes :			244.8	Nos
10. Filter media behind abutment			94	m ³

CALCULATION OF QUANTITY FOR RETAINING WALL

Total length of Retaining Wall =	20	m
Overall height of retaining wall =	5	m
Width of Heel Slab =	2.55	m
Width of Toe Slab =	0.6	m
Thickness of Stem at bottom =	0.6	m
Total width of foundation =	3.75	m
Thickness of foundation near Stem =	0.6	m
Thickness of Heel slab Tip =	0.3	m
Thickness of Toe slab Tip =	0.3	m
Thickness of Stem at top =	0.3	m
Height of Stem =	4.4	m
Thickness of filter media behind retaining wall =	0.6	m
Thickness of levelling course below foundation =	0.15	m
Angle of internal friction for Backfill Material =	32	deg
Average height of Ground Level =	3	m
C/s area of Stem =	1.98	m ²
C/s area of Foundation =	1.7775	m ²
C/s area of filter media behind retaining wall =	2.64	m ²
C/s area of backfilling material=	16.586	m ²
C/s area of Earthwork in Excavation =	12.150	
Volume of concrete in Retaining Wall Stem =	39.6	m ³
Volume of concrete in Retaining Wall Foundation =	35.6	m ³
Total Volume of Concrete in Retaining Wall =	75.2	m ³
Volume of filter media behind retaining wall =	53	m ³
Volume of backfilling material behind wall =	332	m ³
Earth work in filling in front of abutment	36	m ³
Volume of levelling course below foundation =	12.3	m ³
Volume of Earthwork in Excavation =	243.0	m ³
Total number of weep holes =	120	Nos
Total HYSD bar for retaining wall @100=	8.3	Mton
Boulder pitching in front of retaining wall =	12	m ³

SUMMARY OF QUANTITY CALCULATION FOR BRIDGE AT KM - 6.580

				Remarks
6.01	Earthwork in excavation for Soil in ordinary soil	642.85	m ³	Sl. No.1 of Abutment *0.7
	Earthwork in excavation for Soil in hard rock (no blasting)	780.20	m ³	Sl. No.1 of Abutment *0.3+Sl. No. 1 of Pier
	Earth work in Filling =	36	m ³	
6.02	PCC Levelling Course (M15)	66.70	m ³	Sl. No.5 of Abutment +Sl. No. 5 of Pier
6.03	RCC in Open Foundation =	467.55	m ³	Sl. No.3 of Abutment +Sl. No. 3 of Pier
6.04	RCC in Substructure (M35)=	402.2	m ³	Sl. No.4 of Abutment +Sl. No. 4 of Pier
6.05	RCC in Superstructure (M35)=	303	m ³	Based on Design
HYSD bar reinforcement complete as per drawings and technical specifications:				
	Abutment =	48.3	MTon	Sl. No.7 of Abutment
	Pier =	56.0	MTon	Sl. No.6 of Pier
6.06	Retaining wall =	8.3	MTon	
	Crash Barrier =	2.6	MTon	Sl. No.4 of Crash Barrier
	Superstructure=	51.3	MTon	Based on Design
	Total =	166.5	MTon	
6.08	65 mm thick wearing coat	315	sqm	Based on Carriageway and span
6.09	Wearing coat on Footpath	5	m ³	
6.10	Free bearings	4	Nos.	Based on Design
	Pin Bearings	1	Nos.	
	Guided Bearing	1	Nos.	
6.11	Expansion joint (80 mm movement)	39	m	Based on Design
6.12	Helical Void Former	108	m	Based on Design
6.13	Drainage Spout	31	Nos	Based on Design
6.14	Approach Slab	32	m ³	Sl. No.6 of Abutment
6.15	Hand Railing at the edge of Footpath	74	m	Sl. No 5 of Crash Barrier
6.16	Crash Barrier	26	m ³	Sl. No.3 of Crash Barrier
6.17	Earth work in filling	0.0	m ³	
6.18	Weep holes	364.8	Nos.	Sl. No.9 of Abutment
6.19	Backfilling behind Abutment	1105	m ³	Sl. No.2 of Abutment
6.20	Filter media behind abutment	146	m ³	Sl. No.10 of Abutment
6.21	Boulder Grouted in Annular Space	58.5	m ³	Based on Design and Footing Size
6.22	M15 Grouting in Annular Space	98.0	m ³	Based on Design and Footing Size
6.23	Preparation of rock foundation	264	m ²	Based on Footing Size
6.24	Lettering and No. of Bridges	100	Nos	Based on No. of letters
6.25	Providing and applying 2 coats of water based cement paint	740	m ²	Based on Crash Barrier and Railing length

SUMMARY FOR QUANTITY CALCULATION FOR BRIDGE AT KM - 10.418

Item No.	Description	Unit	Value
6.01	for soft rock	cum	1537.01
	for hard rock with no blast	cum	658.72
6.02	PCC Levelling Course (M15)	cum	65.91
6.03	RCC/PCC in open foundation	cum	444.25
6.04	RCC/PCC in Substructure	cum	562.50
6.05	RCC in Superstructure	cum	79.63
6.06	Steel Reinforcement	MT	132.70
<u>HYSD bar reinforcement complete as per drawings and technical specifications:</u>			
	Abutment =	MTon	107.4
	Pier =	MTon	0.0
	Retaining wall =	MTon	24.8
	Crash Barrier @100	MTon	0.7
	Superstructure @100 =	MTon	8.0
	Total =	MTon	116.1
6.08	Bituminous Wearing coat	sqm	128
6.09	Wearing coat on footpath	cum	2
6.10	Tar paper bearing	sqm	37.24
6.11	Expansion joints	m	26.00
6.13	Drainage Spouts	Nos	12.00
6.14	Approach Slab	cum	31.85
6.15	Length of hand rails	Rm	30.00
6.16	Crash Barrier	cum	7.18
6.17	earthwork in filling	cum	0.00
6.18	Weep holes	Nos	344.40
6.19	Backfilling behind earth retaining structures	cum	3271.36
6.20	Filter media behind earth retaining structures	cum	257.46
6.21	Boulder grouted with cement mortar	cum	157.95
6.22	PCC M-15 in annular space	cum	36.654
6.23	Preparation of rock foundation	sqm	242.50
6.24	Lettering and No. of bridges	Nos	40.00
6.25	Coats of cement paint on crash barrier and railing	sqm	300.00

Summary of Gabion Wall Data			
Cl. Ht (m)	X Area (m ²)	Excavation (m ³)	Backfill (m ³)
0.0	0	0	0
0.5	1.50	0.57	0.29
1.0	2.75	2.79	1.15
1.5	4.13	6.38	2.60
2.0	4.625	7.65	3.54
2.5	9.25	12.36	4.62
3.0	9.75	14.00	5.85
3.5	12.75	17.82	7.22
4.0	13.25	19.58	9.48
4.5	16.75	24.02	11.64
5.0	17.25	25.90	14.95
5.5	21.25	30.96	17.65
6.0	21.75	32.95	21.49
6.5	26.25	38.63	24.73
7.0	26.75	40.75	29.11
7.5	32.25	51.30	36.13
8.0	32.75	53.78	41.30

Station	Type of Retaining Structure	Retain Ht.	Retain Length	Culvert Position	X-Area of GW (m ²)	Surface Area of RE Wall (m ²)	Volume of GW (m ³)	RE Wall Actually Provided (m ²)	Addl Qty GW in place replaced RE Wall (m ³)	Deducted Qty GW at Culvert location (m ³)	Exca. Qty for GW (m ³)	Backfill Qty for GW (m ³)	Exca. Qty for REW (m ³)	Backfill Qty for REW (m ³)
1146.065	GW	4.0	0.10		13.25	0.00	1.33				1.96	0.95	0.00	0.00
1150	GW	4.0	3.93		13.25	0.00	52.14				77.04	37.31	0.00	0.00
1160	GW	4.5	10.00		16.75	0.00	167.50				240.20	116.42	0.00	0.00
1160.965	GW	4.5	0.96		16.75	0.00	16.16				23.18	11.23	0.00	0.00
1170	GW	4.5	9.04		16.75	0.00	151.34				217.02	105.19	0.00	0.00
1170.884	GW	4.5	0.88		16.75	0.00	14.81				21.23	10.29	0.00	0.00
1180	GW	5.5	9.12		21.25	0.00	193.72				282.21	160.85	0.00	0.00
1190	GW	5.0	10.00	C	17.25	0.00	172.50			138.00	258.96	149.47	0.00	0.00
1190.887	GW	5.0	0.89	C	17.25	0.00	15.30			138.00	22.97	13.26	0.00	0.00
1191.816	GW	5.0	0.93	C	17.25	0.00	16.03			138.00	24.06	13.89	0.00	0.00
1200	GW	4.5	8.18		16.75	0.00	137.08				196.58	95.28	0.00	0.00
1209.411	GW	3.5	9.41		12.75	0.00	119.99				167.70	67.92	0.00	0.00
1210	GW	3.5	0.59		12.75	0.00	7.51				10.50	4.25	0.00	0.00
1213.667	GW	3.0	3.67		9.75	0.00	35.75				51.33	21.44	0.00	0.00
1220	GW	3.0	6.33		9.75	0.00	61.75				88.64	37.02	0.00	0.00
1224.411	GW	4.0	4.41		13.25	0.00	58.45				86.36	41.83	0.00	0.00
1278.667	GW	4.5	4.26		16.75	0.00	71.29				102.23	49.55	0.00	0.00
1230	GW	5.0	1.33		17.25	0.00	22.99				34.52	19.92	0.00	0.00
1236.13	GW	5.5	6.13		21.25	0.00	130.26				189.77	108.16	0.00	0.00
1237.911	GW	6.0	1.78		21.75	0.00	38.74				58.69	38.27	0.00	0.00
1240	RE	6.5	2.09		0.00	8.00	0.00	8.00	0.00		0.00	0.00	45.42	94.26
1242.167	RE	6.5	2.17		0.00	8.00	0.00	8.00	0.00		0.00	0.00	47.12	97.78
1250	RE	6.5	7.83		0.00	8.00	0.00	8.00	0.00		0.00	0.00	170.32	353.42
1255.715	RE	6.5	5.71		0.00	8.00	0.00	8.00	0.00		0.00	0.00	124.27	257.86
1260	RE	6.5	4.29		0.00	8.00	0.00	8.00	0.00		0.00	0.00	93.17	193.34
1269.215	RE	6.5	9.21		0.00	8.00	0.00	8.00	0.00		0.00	0.00	200.37	415.78
1270	RE	6.5	0.79		0.00	8.00	0.00	8.00	0.00		0.00	0.00	17.07	35.42
1280	GW	6.0	10.00		21.75	0.00	217.50				329.54	214.88	0.00	0.00
1281.374	GW	6.0	1.37		21.75	0.00	29.88				45.28	29.52	0.00	0.00
1284.215	GW	6.0	2.84		21.75	0.00	61.79				93.62	61.05	0.00	0.00
1290	GW	6.0	5.79		21.75	0.00	125.82				190.64	124.31	0.00	0.00
1300	GW	5.0	10.00		17.25	0.00	172.50				258.96	149.47	0.00	0.00
1310	GW	4.0	10.00		13.25	0.00	132.50				195.77	94.82	0.00	0.00
1311.205	GW	4.0	1.20		13.25	0.00	15.97				23.59	11.43	0.00	0.00
1320	GW	4.0	8.80		13.25	0.00	116.53				172.18	83.40	0.00	0.00
1330	GW	4.0	10.00	C	13.25	0.00	132.50			106.00	195.77	94.82	0.00	0.00
1338.195	GW	4.0	8.19		13.25	0.00	108.58				160.44	77.71	0.00	0.00
1340	GW	4.0	1.81		13.25	0.00	23.92				35.34	17.12	0.00	0.00
1350	GW	3.5	10.00		12.75	0.00	127.50				178.20	72.17	0.00	0.00
1353.195	GW	3.5	3.19		12.75	0.00	40.74				56.93	23.06	0.00	0.00
1360	GW	4.5	6.81		16.75	0.00	113.98				163.45	79.23	0.00	0.00
1366.695	GW	5.0	6.69		17.25	0.00	115.49				173.38	100.07	0.00	0.00
1370	GW	5.0	3.31		17.25	0.00	57.01				85.59	49.40	0.00	0.00
1374.671	GW	5.0	4.67		17.25	0.00	80.57				120.96	69.82	0.00	0.00
1380	GW	5.5	5.33		21.25	0.00	113.24				164.97	94.03	0.00	0.00
1390	GW	5.5	10.00		21.25	0.00	212.50				309.58	176.45	0.00	0.00
1400	GW	5.0	10.00		17.25	0.00	172.50				258.96	149.47	0.00	0.00
1410	GW	5.0	10.00		17.25	0.00	172.50				258.96	149.47	0.00	0.00
1413.786	GW	5.0	3.79		17.25	0.00	65.31				98.04	56.59	0.00	0.00
1413.887	GW	5.0	0.10		17.25	0.00	1.74				2.62	1.51	0.00	0.00
1420	GW	4.5	6.11		16.75	0.00	102.39				146.83	71.17	0.00	0.00
1428.886	GW	4.0	8.89		13.25	0.00	117.74				173.96	84.26	0.00	0.00
1430	GW	4.0	1.11		13.25	0.00	14.76				21.81	10.56	0.00	0.00
1430.275	GW	4.0	0.28		13.25	0.00	3.64				5.38	2.61	0.00	0.00
1430.632	GW	4.0	0.36		13.25	0.00	4.73				6.99	3.39	0.00	0.00
1438.132	GW	4.0	7.50		13.25	0.00	99.38				146.83	71.12	0.00	0.00
1440	GW	4.0	1.87		13.25	0.00	24.75				36.57	17.71	0.00	0.00
1450	GW	4.5	10.00		16.75	0.00	167.50				240.20	116.42	0.00	0.00
1453.132	GW	4.5	3.13		16.75	0.00	52.46				75.23	36.46	0.00	0.00
1460	GW	5.0	6.87		17.25	0.00	118.47				177.86	102.65	0.00	0.00
1470	GW	4.5	10.00		16.75	0.00	167.50				240.20	116.42	0.00	0.00
1479.635	GW	5.0	9.63		17.25	0.00	166.20				249.51	144.01	0.00	0.00
1479.635	GW	5.0	0.00		17.25	0.00	0.00				0.00	0.00	0.00	0.00
1480	GW	5.0	0.37		17.25	0.00	6.30				9.45	5.46	0.00	0.00
1486.537	GW	5.5	6.54		21.25	0.00	138.91				202.37	115.35	0.00	0.00
1490	GW	5.5	3.46		21.25	0.00	73.59				107.21	61.10	0.00	0.00
1500	GW	5.5	10.00		21.25	0.00	212.50				309.58	176.45	0.00	0.00
1502.138	GW	5.5	2.14		21.25	0.00	45.43				66.19	37.73	0.00	0.00
1510	GW	5.5	7.86		21.25	0.00	167.07				243.39	138.73	0.00	0.00
1517.138	GW	5.5	7.14		21.25	0.00	151.68				220.98	125.95	0.00	0.00
1520	GW	5.5	2.86		21.25	0.00	60.82				88.60	50.50	0.00	0.00
1530	GW	5.0	10.00		17.25	0.00	172.50				258.96	149.47	0.00	0.00
1530.638	GW	5.0	0.64		17.25	0.00	11.01				16.52	9.54	0.00	0.00
1540	GW	5.0	9.36		17.25	0.00	161.49				242.44	139.93	0.00	0.00

Summary of Unit Rates (Revised) for Changtongya - Longleng Road

Item No.	Description	Unit	Adopted Unit Rate as per MORTH Data Book (Rs)	Item No. in Rate Analysis
1	2	3	4	5
	BILL NO. 8: DRAINAGE AND PROTECTIVE WORKS			
8.01	Lined Random Rubble Masonry Drain on Hill Side, Embankment Toe and Hill Crest Complete as per drawings and Technical Specification Section 300 and 1400 and as directed by the Engineer.	m	4422	
8.02	Covered RCC Rectangular Drain including Reinforcement complete as per drawing and Technical Specification Sections 300, 1000, 1400,1500,1600, 1700 and as directed by Engineer	m	19199	
8.03	Subsurface Drainage complete as per drawings and Technical Specification Section 309			
a	100 mm diameter PVC Perforated Closed Jointed wrapped with Geo-synthetic as per drawings and Technical Specification	m	478	
b	Aggregate Drain below Hill Side Drain as per drawings and Technical Specification	m	401	
8.04	RCC Retaining wall, Gabion Wall			
a	Excavation for RCC retaining wall, Gabion wall as per drawings and Technical Specification, in			
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	261.5	(3.32+3.33)/2
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	427	3.34
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	419	3.9
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	562	3.8 A
b	Back filling behind Earth Retaining Structures Complete as per drawings and Technical Specification Clause 305.	cum	3897	13.9 A
c	Boulder / Concrete Block Pitching on Cut / Embankment Slope complete as per drawings and Technical Specification Section 2500.	cum	4283	15.4
8.05	Gabion Structure for Retaining Earth with segments of mechanically woven (double twist) wire crates filled with boulders including non-woven geotextile as filter media complete as per drawing and Technical Specification Section 700 and 2500	cum	6325	As per Market
8.06	Seeding and Mulching of Cut Slopes / Embankment Slopes complete as per drawings and Technical Specification Clause 308	sqm	996	3.23
8.07	Geo-synthetic Mat for Erosion Control complete as per drawings and Technical Specification clause 706	sqm	1860	As per Market
8.08	Soil Nailing complete as per drawings and Technical Specification Section 3200	m	4893	As per Market
8.09	Temporary Facing complete as per drawings and Technical Specification 3200.	sqm	3045	As per Market
8.10	Reinforced Soil Wall complete as per drawing and Technical Specification Section 3100.			
a)	Facia panels as per Technical Specification Section 3105 including soil reinforcing element, foundation pad, coping beam, all accessories, consumables and components of drainage system (filter media, drainage layer, drain pipe, catch pit etc.), including ground improvement complete.	sqm	7435	7.5 (I) A (type-5) + 7.5(ii) + 12.8 B*0.20
b)	Excavation for RE wall as per drawings and Technical Specification, in			
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	261.5	(3.32+3.33)/2
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	427	3.34
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	419	3.9
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	562	3.8 A
c)	Back filling behind RE Wall Complete as per drawings and Technical Specification Clause 305.	cum	3897	13.9 A
8.11	Construction of Parapet Wall in RR Masonry (CM 1:3) as per Drawings and Technical Specification 1400	cum	10630	13.4 A

Quantity Calculation for Traffic Signs

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS															
			0KM-1KM	1KM-2KM	2KM-3KM	3KM-4KM	4KM-5KM	5KM-6KM	6KM-7KM	7KM-8KM	8KM-9KM	9KM-10KM	10KM-11KM	11KM-12KM	12KM-13KM	13KM-14KM	14KM-15KM	KM15-KM16
	STOP	14.01	2															
	GIVE WAY	14.02	2															
	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03																
	NO STOPPING OR STANDING	14.28																
	NO PARKING	14.29																
	MAXIMUM SPEED LIMIT	14.37	2				2	2	2			2						
	COMPULSORY KEEP LEFT	14.48	4															2
	COMPULSORY KEEP RIGHT	14.49																
	COMPULSORY SOUND HORN	14.57	2							2								
	LEFT HAND CURVE	15.01	1		1			2	1	1	2	1		1	2	1		2
	RIGHT HAND CURVE	15.02	1		1			1	1	1	2	1		1	2	1		
	RIGHT HAIRPIN BEND	15.03																
	LEFT HAIRPIN BEND	15.04					1	1										
	RIGHT HAIRPIN BEND	15.04A					1	1										
	RIGHT REVERSE BEND	15.05	1				1	1	1	1	3	2						1
	LEFT REVERSE BEND	15.06	1				1	1	1	1	3	2						1
	SERIES OF BEND	15.07					2	2			2							2
	SIDE ROAD RIGHT	15.09																
	SIDE ROAD LEFT	15.10																
	Y-INTERSECTION LEFT	15.11																
	Y-INTERSECTION RIGHT	15.12																
	Y-INTERSECTION	15.13																
	X-ROAD	15.14																
	ROUNDABOUT	15.15																
	T-INTERSECTION	15.17																
	T-INT. MAJOR ROAD	15.18																
	STAGGERED INTERSECTION RIGHT	15.20																
	STAGGERED INTERSECTION LEFT	15.21																
	NARROW BRIDGE AHEAD	15.25																
	STEEP ASCENT	15.26		1						1	1	1	1		1	1	1	
	STEEP DESCENT	15.27		1						1	1	1	1		1	1	1	
	PEDESTRIAN CROSSING	15.33																
	SCHOOL AHEAD	15.34																
	BUILT UP AREA	15.35	2															
	PLAYGROUND AHEAD	15.58																
	BARRIER	15.60																
	FALLING ROCKS	15.65																
	SINGLE CHEVRON (SMALL)	15.72	12	12	16	24	24	27	30	27	32	25	16	8	10	10	13	16
	DOUBLE CHEVRON	15.74																
	OBJECT MARKER(LEFT)	15.76	11	10	6	5	4	6	4	6	7	5	4	6	4	5	6	5
	OBJECT MARKER(RIGHT)	15.77	11	10	6	5	4	6	4	6	7	5	4	6	4	5	6	5
	TWO WAY HAZARD MARKER	15.78																
	MAP TYPE ADVANCE DIRECTION SIGN	16.02	4															
	REASSURANCE SIGN	16.05																
	FLAG TYPE DIRECTION SIGN	16.04	2															
	PLACE IDENTIFICATION SIGN	16.06	2															
	TRUCK LAY- BY	16.07																
	WEIGH BRIDGE AHEAD	16.09																
	GANTORY BOARD	16.10	1															
	TOILET	17.05																
	FILLING STATION	17.06																
	HOSPITAL	17.07																
	POLICE STATION	17.13																
	PICNIC SITE	17.14																
	HOME ZONE	17.21																
	BUSTOP	17.35	2															
	STATE HIGHWAY ROUTE MARKER SIGN	20.01																
	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02	6							2	60	51	26	22	24	24	27	34
	TOTAL NO OF SIGNS		69	32	32	42	42	51	46	49	60	51	26	22	24	24	27	34

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in State of Nagaland under SARDP-NE

RITES Ltd.
Highway Division

Quantity Calculation for Traffic Signs

Annexure -9.01

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS											TOTAL
			KM31-KM32	KM32-KM33	KM33-KM34	KM34-KM35	KM35-KM36	KM36-KM37	KM37-KM38	KM38-KM38.5	KM38.5			
	STOP	14.01										4	16	
	GIVE WAY	14.02											2	
	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03											0	
	NO STOPPING OR STANDING	14.28											0	
	NO PARKING	14.29											0	
	MAXIMUM SPEED LIMIT	14.37				2		2		2		4	34	
	COMPULSORY KEEP LEFT	14.48											4	
	COMPULSORY KEEP RIGHT	14.49											0	
	COMPULSORY SOUND HORN	14.57				2		2		2		4	40	
	LEFT HAND CURVE	15.01	1	1	2	2		2		1		1	31	
	RIGHT HAND CURVE	15.02	1	1	2	2		1		1		1	31	
	RIGHT HAIRPIN BEND	15.03											0	
	LEFT HAIRPIN BEND	15.04				1		1		2		2	9	
	RIGHT HAIRPIN BEND	15.04A				1		1		2		2	9	
	LEFT REVERSE BEND	15.05	3										31	
	RIGHT REVERSE BEND	15.06	3										31	
	SERIES OF BEND	15.07	2		2	2		2		2		2	36	
	SIDE ROAD RIGHT	15.09											7	
	SIDE ROAD LEFT	15.10											2	
	Y-INTERSECTION LEFT	15.11											7	
	Y-INTERSECTION RIGHT	15.12											0	
	Y-INTERSECTION	15.13											0	
	X-ROAD	15.14											0	
	ROUNDABOUT	15.15											0	
	T-INTERSECTION	15.17											0	
	T-INT. MAJOR ROAD	15.18											0	
	STAGGERED INTERSECTION RIGHT	15.20											0	
	STAGGERED INTERSECTION LEFT	15.21											0	
	NARROW BRIDGE AHEAD	15.25											0	
	STEEP ASCENT	15.26											14	
	STEEP DESCENT	15.27	1										14	
	PEDESTRIAN CROSSING	15.33											14	
	SCHOOL AHEAD	15.34								2		4	2	
	BUILT UP AREA	15.35											2	
	PLAYGROUND AHEAD	15.58											0	
	BARRIER	15.60											0	
	FALLING ROCKS	15.65											0	
	SINGLE CHEVRON (SMALL)	15.72	34	22	10	25	17	25	26	9		780	0	
	DOUBLE CHEVRON	15.74											0	
	OBJECT MARKER(LEFT)	15.76	11	4	6	4	5	3	5	2		203	0	
	OBJECT MARKER(RIGHT)	15.77	11	4	6	4	5	3	5	2		203	0	
	TWO WAY HAZARD MARKER	15.78											0	
	MAP TYPE ADVANCE DIRECTION SIGN	16.02											4	
	REASSURANCE SIGN	16.05											0	
	FLAG TYPE DIRECTION SIGN	16.04											16	
	PLACE IDENTIFICATION SIGN	16.06											6	
	TRUCK LAY- BY	16.07											0	
	WEIGH BRIDGE AHEAD	16.09											0	
	GANTORY BOARD	16.10											2	
	TOILET	17.05											0	
	FILLING STATION	17.06											0	
	HOSPITAL	17.07											2	
	POLICE STATION	17.13											0	
	PICNIC SITE	17.14											0	
	HOME ZONE	17.21											0	
	BUSTOP	17.35											6	
	STATE HIGHWAY ROUTE MARKER SIGN	20.01		2						2		2	0	
	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02											26	
	TOTAL NO OF SIGNS		68	34	28	45	37	41	52	50		1582		

Calculation of Quantity for Overhead Gantry

S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/m)	Quantity (Kg)
	(A) Truss							
	80mm dia NB Pipe	m	4	14.10			8.36	471.50
	40mm dia NB Pipe at Side Elevation							
	Vertical Side Pipe	m	28	1.24			3.56	123.60
	Horizontal Pipe	m	4	0.74			3.56	10.54
	Diagonal pipe at ends	m	4	1.56			3.56	22.21
	Diagonal pipe at midway	m	22	1.63			3.56	127.66
	25mm dia NB Pipe at top & bottom Elevation							
	Diagonal pipe at ends	m	4	1.22			2.41	11.76
	Diagonal pipe at midway	m	22	1.30			2.41	68.93
	Cross Pipe	m	24	0.67			2.41	38.75
	(B) Vertical Support of Pipe 300 mm Dia	m	2	5.00			49.30	493.00
							Total wt (A+B)	1367.96
S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/cum)	Quantity
	(C) Gusset Plates							
	Gusset Plate at top of Truss Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Top frame of Truss	Sqm	8	0.56	0.30	0.01	7854	105.56
	Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Base of Truss Frame	Sqm	8	0.56	0.30	0.01	7854	105.56
	M.S. Plates welded to Bolts	Sqm	16	0.10	0.10	0.01	7854	12.57
							Total wt (C)	977.67
	Total weight of One Overhead Gantry (A+B+C) in (kg)							2345.63

Location of Proposed Gantry Signs

S. No.	Location	No. of Gantry Signs
1	Start of Project Road	1
2	End of Project Road	1
	Total	2

S. No.	Description	Unit	No. of Overhead Gantry	No	Length (m)	Breadth (m)	Height / Depth (m)	Quantity	Say
1	Gantry Mounted Overhead Signs	Sqm	2	6	2.30	1.30		35.88	40
2	Foundation Pedestal								
	Excavation	cum	2	2	2.40	2.40	2.10	48.38	50
	PCC M-15 in footing	cum	2	2	2.40	2.40	0.10	2.30	3
	M-25 RCC in Footing	cum	2	2	2.20	2.20	0.30	5.81	6
	M-25 RCC in Pedestal	cum	2	2	1.00	1.00	2.45	9.80	10
	Reinforcement @ 60kg/cum of concrete	tonne	2					1.87	2
3	Overhead Gantry	tonne	2					4.69	5

Quantity Calculation for Pavement Marking

Total Length of road	38500	m
Total Length of Curved portion of Road	18695	m
Total Length of Straight portion of Road	19804	m

Description	Unit	Length of Road (m)	No	Length of Line (m)	Width (m)	Quantity
Lane Marking (St. Portion)	sqm	19804	2201	3.00	0.10	660
Lane Marking (Curved Portion)	sqm	18695	3116	3.00	0.10	935
Edge Marking	sqm	38500	2	38500	0.15	11550
					Total	13145
Directional Arrows & Lettering	No.		25			30
Road studs on Edge /Centre line marking @ 6 m c/c	No.	38500	19250			19250

Calculation of Quantities for Junction & Median Improvement

S No.	Chainage (Km)	Type of Junction	Island in Sq mt	Area Road in Sq mt	E/S Area (Cross Leg) in Sq mt	Length of Kerb on Island and Median (Lm)	No. of Arrows		Type of Junction
							Straight	Straight-Left/Right	
1	0.000	Major	458	6917	650	540	20	5	T-Junction
		Minor		4250	2250				
		Total	458.00	11167.00	2900.00	540.00	20.00	5.00	

SUMMARY OF QUANTITIES							
S No.	Description	unit	No	Area (sqm)	Th. (m)	Qty.	Say
1	BC	cum	1	11167.00	0.030	335.01	335.00
2	DBM	cum	1	11167.00	0.050	558.35	558.00
3	WMM	cum	1	11167.00	0.250	2791.75	2792.00
4	GSB	cum	1	11167.00	0.180	2010.06	2010.00
5	Subgrade	cum	1	14067.00	0.500	7033.50	7034.00
6	Prime Coat	sqm	1	11167.00		11167.00	11167.00
7	Tack Coat	sqm	1	11167.00		11167.00	11167.00
8	Kerb Stone in Island & Medians	m		540.00		540.00	540.00
9	Soil Filling in Medians & Island	cum	1	458.00	0.680	311.44	311.00